QUALIFICATION STANDARD
AND PLAN (QSP)

Weapons Engineering Technician (W Eng Tech)
NQual (AKQO)

MITE ID
Residential Phase – 118943

Regular Force: Weapons Engineering Technician 00366 (W Eng Tech)
Training Authority (TA): Commander Naval Training Group (CNTG)
Curriculum Control Authority (CCA): NFS(P)

MARPAC RDIMS 1376298
Chap 1 & 2 approved DD MM YYYY
Chap 3 & 4 approved DD MM YYYY

RECORD OF AMENDMENTS

1. The following versions of this QSP have been approved for use in support of the conduct of training. Amendments detailed in Annexes D and/or L.

Revision	Change Description	Date	Authority
1.0		21 Feb 11	DMTE 2-4-2
2.0		27 Jun 2014	O.S.B.//Cdr D.J. Benoit CFNES Cmdt
2.1	QSP change to new format Addition of Unit Technical Inspector MITE 302486	19 Sept 24	NTG HQ

NOTE:

Personnel programming courses must ensure that their hard copy of this document is current and identical to the electronic copy on the NTG Website. See above for current changes.

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FOREWORD

1. This QSP is issued on the authority of the Commander of the Royal Canadian Navy and managed by the Commander Naval Training Group (CNTG) on behalf of the Assistant Chief of Naval Staff Personnel and Training (A/CNS P&T).

2. This QSP is based upon the following:

   a. A-P9 – 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

   b. A-PD-055-002/PP-002, Non-commission Member General Specification published 09 February 2011; and

   c. and A-PD-055-002/PP-001, Weapons Engineering Technician 00366 published 01 September 2011;

3. This QSP publication is effective on receipt and supersedes AKQO QSP revision 2.0, dated 27 June 2014. This QSP is designed to assist in developing, conducting, evaluating, and validating the Individual Training and Education (IT&E) required for training.

4. The QS at Chapters 1 and 2 were developed by the Qualification Standard Writing Board (QSWB) convened 12-30 April 2010 at CFFS(E) Bldg N92 Rm 300 working in collaboration with subject matter experts drawn from participating organizations. The subsequent TP at Chapters 3 and 4 were developed by the Training Plan Writing Board (TPWB) convened CFFS(E) Bldg N92 Rm 300 working in collaboration with subject matter experts drawn from participating organizations. All aspects of this document were prepared in accordance with the concept of training outlined in A-P9-050 Series, Canadian Forces Manual of Individual Training and Education. Full details of the WBs are contained within Annex D and L.

5. The criteria for major and minor amendments, and the approval process, are identified in NPTORD 4500-22.

6. In circumstances where the QS and/or TP cannot be adhered to, submission of a QSP Deviation Request is required.

QUALIFICATION STANDARD

MARPAC RDIMS 1376298
Chap 1 & 2 approved DD MM YYYY
Chap 3 & 4 approved DD MM YYYY

CHAPTER 1 QUALIFICATION STANDARD

AIM

1. The aim of this QSP is to establish the requirements and training for the Weapons Engineering Technician (W ENG Tech) to perform their duties as a W ENG Tech Radar System Maintainer. Possession of this qualification will allow personnel to maintain ship's Radar systems throughout the fleet.

GRANTING OF NQUAL

2. A trainee will achieve the qualification AKQO W ENG Tech RADAR Applications when the standards listed in the Performance Objectives (POs) have been met.

a. Radar Applications, MITE 118943

CO-QUALIFICATION

3. In addition to the performance objectives specific to this training, the following additional qualifications are included in this training program to meet the requirements of the job. The MITE and NQuals associated with these qualifications shall be granted upon completion of the training program and shall be listed within the course report and will be delivered IAW the following:

a. DRMIS Mod 2, MITE 119097; and

b. Unit Technical Inspector, MITE 302486.

PREREQUISITES

4. To be selected for this course the loading authority must ensure that trainees meet the following prerequisites:

a. AKNV_ Jr W Eng Tech EQ2 MITE 118939 or MITE 123762; and

b. AKPK_W Eng RQS1 Academics MITE 118966.

CHAPTER 2 PERFORMANCE OBJECTIVES

PO 001 Maintain the Long Range Radar System

1. Performance: Maintain the Long Range Radar System.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) Long Range Radar System / MPT; and

      (7) Assistance.

   a. Denied:

      (1) Nil.

   b. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The W Eng Tech Radar Systems Maintainer shall maintain the Long Range Radar System, IAW references, by:

   a. Describing the Long Range Radar System;

   b. Initializing the Long Range radar System;

   c. Repairing the Antenna System;

   d. Repairing the Aerial Assembly;

   e. Repairing the SMART-S Processing Cabinet; and

   f. Performing preventative maintenance on the Long Range Radar System.

4. References:

   a. A01 – A32.

5. Task Numbers:

   a. D001 – D021.

6. Training Limitations:

   a. Nil.

7. Remarks:

   a. Nil.

PO 002 Maintain the Medium Range Radar

1. Performance: Maintain the Medium Range Radar.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) Radar System / MPT; and

      (7) Assistance.

   b. Denied:

      (1) Nil.

   b. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The W Eng Tech Radar Systems Maintainer shall maintain the Medium Range Radar System, IAW references, by:

   a. Describing the Medium Range Radar System;

   b. Initializing the Medium Range radar System;

   c. Repairing the Medium Range Radar transmitter / receiver;

   d. Repairing the Medium Range Radar signal data processor;

   e. Repairing the Medium Range Radar peripheral equipment, to include:

      (1) PCC;

      (2) AE;

      (3) Cooling Unit;

(4) DAU; and

(5) PDU.

f. Performing preventative maintenance on the Medium Range Radar System.

4. References:

a. A01 – A26;

b. A33 – A39; and

c. C07 – C08.

5. Task Numbers:

a. D001 – D021.

6. Training Limitations:

a. Nil.

7. Remarks:

a. Nil.

PO 003 Maintain Identification Friend or Foe System

1. Performance: Maintain Telephonic Identification Friend or Foe (IFF).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) IFF / MPT; and

      (7) Assistance.

   c. Denied:

      (1) Nil.

   b. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The W Eng Tech Radar Systems Maintainer shall maintain the Telephonic IFF System, IAW references, by:

   a. Describing the IFF;

   b. Initializing the IFF;

   c. Repairing the IFF Transponder Assembly;

   d. Repairing the IFF Interrogator; and

   e. Performing preventative maintenance on the IFF.

4. References:

   a. A01 – A26; and

   b. A40 – A45.

5. Task Numbers:

   a. O001 – O012.

6. Training Limitations:

   a. Nil.

7. Remarks:

   a. Nil.

PO 004 Maintain Electronic Support Measures System

1. Performance: Maintain Electronic Support Measures (ESM) System.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) ESM System / MPT; and

      (7) Assistance.

   d. Denied:

      (1) Nil.

   b. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The W Eng Tech Radar Systems Maintainer shall maintain the ESM System, IAW references, by:

   a. Describing the ESM System;

   b. Initializing the ESM System;

   c. Repairing the ESM Reception Group;

   d. Repairing the ESM Signal Measurement Group;

   e. Repairing the ESM Signal Processing Group; and

   f. Performing preventative maintenance on the ESM System.

4. References:

   a. A01 – A26; and

b. A46 – A49.

5. Task Numbers:

a. E001 – E012.

6. Training Limitations:

a. Nil.

7. Remarks:

a. Nil.

PO 005 Maintain Electronic Support Measures System (Sea Search)

1. Performance: Maintain ESM System (Sea Seach).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) Sea Search / MPT; and

      (7) Assistance.

   e. Denied:

      (1) Nil.

   b. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The W Eng Tech Radar Systems Maintainer shall maintain the ESM System (Sea Search), IAW references, by:

   a. Describing the ESM System (Sea Search);

   b. Initializing the ESM System (Sea Search);

   c. Repairing the ESM System (Sea Search); and

   d. Performing preventative maintenance on the ESM System (Sea Search).

4. References:

   a. A01 – A26; and

   b. C01 – C06.

5. Task Numbers:

a. E001 – E012.

6. Training Limitations:

a. Nil.

7. Remarks:

a. Nil.

PO 006 Maintain Communications Intercept System

1. Performance: Maintain Communications Intercept System.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) Communications Intercept System / MPT; and

      (7) Assistance.

   f. Denied:

      (1) Nil.

   b. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The W Eng Tech Radar Systems Maintainer shall maintain the Communications Intercept System, IAW references, by:

   a. Describing the Communications Intercept System;

   b. Initializing the Communications Intercept System;

   c. Repairing the Communications Intercept System; and

   d. Performing preventative maintenance on the Communications Intercept System.

4. References:

   a. A01 – A26; and

   b. A51 – A56.

5. Task Numbers:

   a. E013 – E029.

6. Training Limitations:

   a. Nil.

7. Remarks:

   a. Nil.

PO 007 Maintain Combat Management System

1. Performance: Maintain Combat Management System (CMS).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) CMS / MPT; and

      (7) Assistance.

   g. Denied:

      (1) Nil.

   b. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The W Eng Tech Radar Systems Maintainer shall maintain the CMS, IAW references, by:

   a. Describing the CMS;

   b. Initializing the CMS;

   c. Repairing the CMS; and

   d. Performing preventative maintenance on the CMS.

4. References:

   a. A01 – A26; and

   b. A57 – A66.

5. Task Numbers:

a. O013 – O020.

6. Training Limitations:

a. Nil.

7. Remarks:

a. Nil.

PO 008 Utilize DRMIS Portal as a Maintenance Supervisor

1. Performance: Utilize DRMIS Portal as a Maintenance Supervisor of the DRMIS IAW PO 002 of QSP 119097.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Computer with LAN access;

      (4) Printer;

      (5) DRMIS account; and

      (6) Access to DRMIS Portal.

   h. Denied:

      (1) Nil.

   b. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: PO 008 shall be carried out IAW PO 002 of the DRMIS QSP 119097.

4. References:

   a. IAW PO 002 of the DRMIS QSP.

5. Task Numbers:

   a. N/A.

6. Training Limitations:

   a. Nil.

7. Remarks:

   a. Nil.

TRAINING PLAN

MARPAC RDIMS #
Chap 1 & 2 approved DD MM YYYY
Chap 3 & 4 approved DD MM YYYY

CHAPTER 3 TRAINING MANAGEMENT DETAILS

TRAINING STRATEGY

1. The following training strategies were selected for this course:

   a. Enabling Checks (ECs) are formative tests that help both Trainee and instructors recognize progress or gaps in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. They also reinforce learning, so that it will be better retained for longer periods. Formative tests are designed to give feedback to both the instructor(s) and Trainees. Each EO must have at least one EC. Marks for ECs are not calculated in the Trainee's final mark;

   b. Performance Checks (PCs) are summative tests used to determine if the Trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Summative tests can take the form of PCs or Critical ECs (reserved for safety related content). Supplemental PCs are administered upon failure of a PC or critical EC and must be in the same format and have the same standard as the original PC; and

   c. Use of Applicable Training Simulator software.

TRAINING DURATION

2. Based on existing resources and approved instructional design requirements, the approved Course Design authorizes 60 training days.

Actual Training Time (ATT)	446 periods
Training Administrative Time (TAT)	26 periods
Base Administrative Time (BAT)	7 periods
Training Travel Time (TTT)	0.00
Course Duration (Total Calendar Days) (ATT+TAT+BAT+TTT)/8	60 training days

REQUIRED COURSE STAFFING

3. The following key personnel are required to conduct the training.

Course Director				
Rank	MOSID	Qualifications Required	Qty	PO / EO

Instructors				
Rank	MOSID	Qualifications Required	Qty	PO / EO
MS/PO2	00366	W ENG Tech Radar Applications Instructor Training (IT)	2	
PO1	00366	W ENG Tech Maintenance Manager	1	

Guest Lecturers/Others (May be civilian)				
Lecturer	# Instructor	Subject	EO #	# period

TRAINING LIMITATION

4. A training limitation may be due to constraints of resources, personnel or equipment. A training limitation is the difference between what a qualified individual must be able to do and what training can prepare them to do. The following limitations are identified below:

a. Limitation 1; and

b. Limitation 2.

COURSE CAPACITY

5. The capacity per session is based on the guidelines in CFITES Vol 5 – Instructor to Learner Ratios:

a. Optimum – 1:6;

b. Maximum Load – 1:8; and

c. Minimum – 1:2.

DESIGNATED SCHEDULING AND COURSE LOADING

6. Training will be scheduled utilizing the HRMS Training (MITE) module to meet the training requirements (IAW NPTORD 4540-2).

7. The authority to load members onto this course will be IAW NPTORD 4540-43. NTG HQ is to be included as an info addressee on all loading messages. Course loading will be accomplished by:

a. The Career Manager for career courses; or

b. NFS Campus Support for occupation and specialty courses.

QUALIFICATION AND ASSESSMENT

8. This is a pass/fail course. Students must successfully complete all POs to achieve the qualification.

9. Students will be monitored and assessed both on performance and conduct throughout the course. Counselling, progress monitoring, and maintenance of student files will be in accordance with NPTORD 4500-22. Although not mandated specifically in the POs, the general demands of service members and the leadership demands of junior and senior leaders form an integral part of individual training. As such, students are presumed to possess the leader and/or follower attributes commensurate with their rank.

UNSATISFACTORY PROGRESS

10. Any student who fails to successfully complete all POs in the allocated number of attempts will be referred to the Commandant (Cmdt) or Commanding Officer (CO) of the training establishment (TE) to determine the requirement for a TRB IAW NPTORD 4500-54.

REMOVAL FROM TRAINING

11. A TE CO/Cmdt may direct that a student be removed from training for reasons of deficiency in performance, conduct (discipline, attitude, teamwork, safety, leadership), missed training, medical, and other non-performance/conduct related issues. In these cases, Cease Training Procedures are to be strictly adhered to IAW NPTORD 4500-29 or 4500-54.

RECERTIFICATION REQUIREMENTS

12. There are no additional requirements to maintain currency in this qualification.

LANGUAGE OF INSTRUCTION

13. The language of instruction shall be in accordance with the Delivery of Training and Education in Both Official Languages (DAOD 5039-6).

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

14. PLARs shall be conducted in accordance with DAOD 5031-1 Canadian Forces Military Equivalencies Program, through the RCN PLAR submissions coordinator.

POST COURSE REVIEWS (PCR)

15. PCRs are to be conducted on all steady state courses IAW NPTORD 4500-23.

COURSE REPORTING

16. Formal Course: A Canadian Forces Course Report, Form CF 377 and/or CF289 Certificate of Military Achievement, shall be prepared and distributed in accordance with DAOD 5031-9. The Course Report shall be descriptive, identifying the student's strengths and weaknesses with specific comments on any PO for which the student demonstrated outstanding performance or had difficulty achieving the standard, any conduct relevant to course performance, and any conditions or situations that affected course performance.

CHAPTER 4 LESSON SPECIFICATIONS

PO 001 – Maintain the Long Range Radar System

EO 001.01 Describe the Long Range Radar System

1. Performance: Describe the Long Range Radar System

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) DWAN;

        (4) Tools and test equipment;

        (5) PPE;

        (6) Long Range Radar System / MPT; and

        (7) Assistance.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The trainee shall describe the Long Range Radar System, IAW references, by:

    a. System overview;

    b. Capabilities and limitations;

    c. Theory of operation;

    d. Power and environmental distribution; and

    e. Controls and indicators.

4. Time Required:

a. EO = 18 x 50 min periods;

5. Instructional Methods:

a. Interactive lecture;

b. Demonstration;

c. Trainee practice; and

d. PC.

6. Training Aids:

a. MPT or Radar;

b. References;

c. Audio visual aids; and

d. Handouts.

7. Test Details:

a. EO 001.01 will be tested as part of PC 001 IAW Annex H.

8. References:

a. A01 – A32.

9. Remarks:

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	System Overview to include: 1. Major Components:	L	2	

	a. Antenna system; b. Junction box; c. Person aloft switch; d. SMART-S Processing Cabinet (SPC); e. Drive Control and cooling cabinet; f. Air dryer; g. Maintainer Terminal; and h. External interfaces. 2. Location on the Ship.			
TP2	Capabilities and limitations to include: 1. Detection range; 2. Plot accuracies; 3. Coverage diagrams; and 4. Performance on equipment level; Antenna system parameters to include: 1. Rotational speeds; 2. Pitch and roll limits; 3. RF output; 4. Emission modes; 5. Dimensions and weight; and 6. Environmental limits Sub-components to include: 1. SPC; 2. ESM blanking pulse; and 3. Air dryer.			

TP3	Theory of Operation to include: 1. Operational functions, to include: a. Air surveillance; b. Surface surveillance; c. Jammer surveillance; d. Surface Gun Fire control; and e. Operational support; 2. Functional chains, to include: a. Antenna Assembly, to include: (1) Antenna Array pack; (2) TR-Books; (3) Multi-channel receiver; (4) Radar Signal generator; (5) Infra function; (6) Rotary joint unit; and (7) Drive assembly. b. Signal Processing Cabinet, to include: (1) Signal Processing Unit, to include: (a) Form Beams; (b) Form Pulses;	L	3	

		(c) Form Doppler channels; (d) Form hits; (e) Provide FC video; (f) Determine clutter range; (g) Provide ducting measurement; (h) Calculate power samples; (i) Measure power samples; (j) Form display video; and (k) Measure RX test pulse. (2) Data Processing Unit, to include: (a) Extraction functions; (b) Tracking functions; (c) Fire control functions; (d) Duct control functions; and (e) DPU control.			

	(3) Gateway BIT and control; and (4) Infrastructure. 3. States & modes, to include: a. Peace / War mode selection; b. Radar mode selection, to include: (1) Defence mode; and (2) Surveillance mode; c. System control selection, to include: (1) Remote; and (2) Local. d. System Readiness states, to include: (1) Isolated; (2) Off; (3) Start-up; (4) Standby; (5) Operational; (6) Dormant; and (7) Prepare for shutdown. 4. Subsystem states and modes, to include: a. Antenna subsystem, to include:			

	(1) Antenna subsystem modes; (2) Antenna subsystem states; (3) Antenna system function states, to include: (a) Antenna climate control; (b) Antenna Rotation guard; and (c) Transmitter. b. Signal processing cabinet.			
TP4	Power and environmental distribution, to include: 1. Power distribution; 2. Cooling distribution; and 3. Air distribution.	L	1	
TP5	Controls and indicators, to include: 1. Antenna Assembly; 2. SMART-S Processing Cabinet; 3. Direct Drive and Cooling Cabinet; 4. Air Dryer; and 5. Person aloft switch.	L	2	

PO 001 Maintain the Long Range Radar System

EO 001.02 Initialize the Long Range Radar System

1. Performance: Initialize the Long Range Radar System

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) DWAN;

        (4) Tools and test equipment;

        (5) PPE;

        (6) Long Range Radar System / MPT; and

        (7) Assistance.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The trainee shall initialize the Long Range Radar System, IAW references, by:

    a. Flash up / Shut down process;

    b. Demonstrate and practice Flash up / Shut down;

    c. Maintainer Terminal Functions; and

    d. Demonstrate and practice Maintainer Terminal Functions.

4. Time Required:

    a. EO = 7 x 50 min periods;

5. Instructional Methods:

a. Interactive lecture;

b. Demonstration;

c. Trainee practice; and

d. PC.

6. Training Aids:

a. MPT or Radar;

b. References;

c. Audio visual aids; and

d. Handouts.

7. Test Details:

a. EO 001.02 will be tested as part of PC 001 IAW Annex H.

8. References:

a. A01 – A32.

9. Remarks:

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Flash up / Shutdown process: 1. System pre-operation checks, to include: a. Perform the SPC check;	L	1	

	b. Set the Person aloft switch to OFF; c. Perform the radar antenna system check; d. Set the person aloft switch to SAFE TO OPERATE; e. Perform the DCCC check; and f. Perform the air dryer check; 2. System flash-up checks, to include: a. Infra on; b. System on; c. Default state settings; d. Set to operational; e. Availability monitoring and control; and f. Power dip. 3. System shut-down checks, to include: a. System off; and b. Infra off. 4. System post-operations, to include: a. Post-operation procedures SPC; b. Post-operation procedures DCCC; and c. Post-operation procedures air dryer.			

TP2	Demonstrate and practice Flash up / Shut down.	D P	0.5 0.5	
TP3	Maintainer Terminal functions, to include: 1. Maintainer terminal setup; 2. Maintainer terminal page overview; 3. Changes system settings; 4. Control page; 5. System parameters; 6. Display and plot control; 7. Safety; 8. Monitoring; 9. System BIT; and 10. Test facilities.	L D P	0.5 0.5 0.5	
TP4	Demonstrate and practice Maintainer Terminal Functions.	L	2	

PO 001 Maintain the Long Range Radar System

EO 001.03 Repair the Antenna System

1. Performance: Initialize the Long Range Radar System

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) DWAN;

        (4) Tools and test equipment;

        (5) PPE;

        (6) Long Range Radar System / MPT; and

        (7) Assistance.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The trainee shall repair the Long Range Radar System, IAW references, by:

    a. Localizing fault to LRU;

    b. Proper removal and replacement of LRU; and

    c. Verify operational readiness.

4. Time Required:

    a. EO = 35 x 50 min periods; and

    b. EC = 8 x 50 min periods.

5. Instructional Methods:

a. Lecture;

b. Demonstration;

c. Trainee practice; and

d. EC / PC.

6. Training Aids:

a. MPT or Radar;

b. References;

c. Audio visual aids; and

d. Handouts.

7. Test Details:

a. EO 001.03 will be tested in EC 001.03 and as part of PC 001 IAW Annex H; and

b. EO 001.03 will have a 50 min time limit (corrective measures start at minute 45).

8. References:

a. A01 – A32.

9. Remarks:

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Localize fault to LRU, to include:	D	0.33	

	1. Reviewing BIT reporting page; 2. Recognizing faults on maintainer terminal subpages; and 3. Interpret diagrams.	P	46	
TP2	Proper removal and replacement of LRU.	D P	0.33 15	
TP3	Verify operational readiness.	D P	0.33 6	

PO 001 Maintain the Long Range Radar System

EO 001.04 Repair the Aerial Assembly

1. Performance: Repair the aerial assembly.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) Long Range Radar System / MPT; and

      (7) Assistance.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The trainee shall repair the aerial assembly, IAW references, by:

   a. Explain the aerial assembly, to include:

      (1) Shock absorbing foundation;

      (2) Drive assembly;

      (3) Rotary joint unit;

      (4) Junction box; and

      (5) Locking device.

   b. Localize fault to LRU;

c. Proper removal and replacement of LRU; and

d. Verify operational readiness.

4. Time Required:

a. EO = 27 x 50 min periods; and

b. EC = 1 x 50 min periods.

5. Instructional Methods:

a. Lecture;

b. Demonstration;

c. Trainee practice; and

d. EC / PC.

6. Training Aids:

a. MPT or Radar;

b. References;

c. Audio visual aids; and

d. Handouts.

7. Test Details:

a. EO 001.04 will be tested in EC 001.04 as part of PC 001 IAW Annex H.

8. References:

a. A01 – A32.

9. Remarks:

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Explain the aerial assembly, to include: 1. Shock Absorbing Foundation; 2. Drive assembly; 3. Rotary Joint Unit; 4. Junction Box; and 5. Locking Device.	L	5	
TP2	Localize fault to LRU.	D P	1 12	
TP3	Proper removal and replacement of LRU.	D P	1 5	
TP4	Verify operational readiness.	D P	1 2	

PO 001 Maintain the Long Range Radar System

EO 001.05 Repair the Drive Control & Cooling Cabinet (DCCC) and Air Dryer

1. Performance: Repair the DCCC and air dryer.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) Long Range Radar System / MPT; and

      (7) Assistance.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The trainee shall repair the DCCC and air dryer, IAW references, by:

   a. Explaining major components of the DCCC;

   b. Localizing fault to LRU;

   c. Removal and replacement of LRU; and

   d. Verifying operational readiness.

4. Time Required:

   a. EO = 19 x 50 min periods; and

   b. EC = 8 x 50 min periods.

5. Instructional Methods:

   a. Lecture;

   b. Demonstration;

   c. Trainee practice; and

   d. EC / PC.

6. Training Aids:

   a. MPT or Radar;

   b. References;

   c. Audio visual aids; and

   d. Handouts.

7. Test Details:

   a. EO 001.05 will be tested in EC 001.05 as part of PC 001 IAW Annex H;

   b. EC 001.05 will have a 50 minute time limit (corrective measures start at minute 45).

8. References:

   a. A01 – A32.

9. Remarks:

   a. MPT may be utilized;

   b. Projects and exercises may be assigned; and

   c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

   a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Explain major components of the DCCC and air dryer to include: 1. Drive Control Unit to include: a. Power; b. Indicators; c. Commands; and d. Status. 2. Coolant Unit to include: a. Power; b. Indicators; c. Commands; and d. Status. 3. Air dryer to include: a. Power; b. Indicators; and c. Controls.	L	5	
TP2	Localize fault to LRU.	D P	1 10	
TP3	Proper removal and replacement of LRU.	D P	0.5 1	
TP4	Verify operational readiness.	D P	0.5 1	

PO 001 Maintain the Long Range Radar System

EO 001.06 Repair the SMART-S Processing Cabinet

1. Performance: Repair the SMART-S Processing Cabinet.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) Long Range Radar System / MPT; and

      (7) Assistance.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The trainee shall repair the SMART – S processing cabinet, IAW references, by:

   a. Explaining features of the SMART-S processing cabinet;

   b. Localizing fault to LRU;

   c. Removal and replacement of LRU; and

   d. Verifying operational readiness.

4. Time Required:

   a. EO = 17 x 50 min periods; and

   b. EC = 1 x 50 min periods; and

c. PC = 12 x 50 min periods.

5. Instructional Methods:

a. Lecture;

b. Demonstration;

c. Trainee practice; and

d. EC / PC.

6. Training Aids:

a. MPT or Radar;

b. References;

c. Audio visual aids; and

d. Handouts.

7. Test Details:

a. EO 001.06 will be tested in EC 001.06 as part of PC 001 IAW Annex H;

b. PC 001 will have a 50 minute time limit (corrective measures start at minute 45).

8. References:

a. A01 – A32.

9. Remarks:

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Features of the SMART-S Processing Cabinet (SPC) to include: 1. Signal Processing Unit (SPU): a. Beam forming; and b. Video processing. 2. Data Processing Unit (DPU): a. Plot forming; and b. Tracking. 3. Gateway, BIT and Control (GBC): a. User adaptation; b. Central BIT; and c. Timing and control. 4. Infrastructure (Infra); 5. Power management; 6. Environmental management; 7. Network switch; and 8. Fibre multiplexer.	L	3	
TP2	Localize fault to LRU.	D P	1 12	
TP3	Proper removal and replacement of LRU.	D P	0.5 1	
TP4	Verify operational readiness.	D P	0.5 1	

PO 001 Maintain the Long Range Radar System

EO 001.07 Perform Preventative Maintenance (PM) on the Long Range Radar System

1. Performance: Perform PM on the Long Range Radar System.

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) DWAN;

        (4) Tools and test equipment;

        (5) PPE;

        (6) Long Range Radar System / MPT; and

        (7) Assistance.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The trainee shall perform PM on the Long Range Radar System, IAW references, by:

    a. Explaining content and use of PM routines (Nys, NKs, and NTs);

    b. Describe the various types of PM routines;

    c. Explain a complex procedure utilizing the PM manual; and

    d. Demonstrate / practice a PM routine.

4. Time Required:

    a. EO = 9 x 50 min periods.

5. Instructional Methods:

   a. Lecture;

   b. Demonstration; and

   c. Trainee practice.

6. Training Aids:

   a. MPT or Radar;

   b. References;

   c. Audio visual aids; and

   d. Handouts.

7. Test Details:

   a. EO 001.07 will not be tested.

8. References:

   a. A01 – A32.

9. Remarks:

   a. MPT may be utilized;

   b. Projects and exercises may be assigned; and

   c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

   a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Explain content and use of PM routines (NYs, NKs, and NTs).	L	2	
TP2	Describe the various types of PM routines.	L	1	

TP3	Explain a complex procedure utilizing the PM manual	L	2	
TP4	Demonstrate / practice a PM routine	D P	1 3	

PO 002 – Maintain the Medium Range Radar System

EO 002.01 Describe the Medium Range Radar

1. Performance: Describe the Medium Range Radar System

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) DWAN;

        (4) Tools and test equipment;

        (5) PPE;

        (6) Radar / MPT; and

        (7) Assistance.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The trainee shall describe the Medium Range Radar System, IAW references, by:

    a. Describe the Medium Range Radar System, to include:

        (1) System block diagram.

    b. Capabilities and limitations;

    c. Theory of operation; and

    d. Controls and indicators.

4. Time Required:

    a. EO = 8 x 50 min periods;

5. Instructional Methods:

   a. Interactive lecture;

   b. Demonstration;

   c. Trainee practice; and

   d. EC.

6. Training Aids:

   a. MPT or Radar;

   b. References;

   c. Audio visual aids; and

   d. Handouts.

7. Test Details:

   a. EO 002.01 will be tested in EC 002.01 as part of PC 002 IAW Annex H.

8. References:

   a. A01 – A26;

   b. A33 – A39; and

   c. C07 – C08.

9. Remarks:

   a. MPT may be utilized;

   b. Projects and exercises may be assigned; and

   c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

   a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Describe the Medium Range Radar System, to include: 1. System block diagram	L	0.5	
TP2	Capabilities and limitations.	L	2.5	
TP3	Theory of Operation.	L	2.5	
TP5	Controls and indicators.	L D	2.5	

PO 002 Maintain the Medium Range Radar System

EO 002.02 Initialize the Medium Range Radar System

1. Performance: Initialize the Medium Range Radar System

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) DWAN;

        (4) Tools and test equipment;

        (5) PPE;

        (6) Radar / MPT; and

        (7) Assistance.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The trainee shall initialize the Medium Range Radar System, IAW references, by:

    a. Explain Flash up / Shut down process;

    b. Load / verify the software; and

    c. Carry out system functional checks and verify operational readiness

4. Time Required:

    a. EO = 3 x 50 min periods;

5. Instructional Methods:

    a. Interactive lecture;

b. Demonstration;

c. Trainee practice; and

d. EC.

6. Training Aids:

a. MPT or Radar;

b. References;

c. Audio visual aids; and

d. Handouts.

7. Test Details:

a. EO 002.02 will be tested in EC 002.02 as part of PC 002 IAW Annex H.

8. References:

a. A01 – A26;

b. A33 – A39; and

c. C07 – C08.

9. Remarks:

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Explain flash-up / shutdown process, to include: 1. System pre-operation procedure;	L P	0.5 0.5	

	2. System flash-up warnings; 3. System flash-up procedure; 4. System shut down procedure; and 5. System post operation procedure.			
TP2	Load / verify software.	D P	0.5 0.5	
TP3	Carry out system functional checks and verify operational readiness.	L P	0.5 0.5	

PO 002 Maintain the Medium Range Radar System

EO 002.03 Repair the Medium Range Radar TX / RX

1. Performance: Repair the Medium Range Radar TX / RX.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) Radar / MPT; and

      (7) Assistance.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The trainee shall repair the Medium Range Radar TX / RX, IAW references, by:

   a. Explain the components in the Medium Range Radar TX / RX;

   b. Localizing fault to LRU;

   c. Proper removal and replacement of LRU; and

   d. Verifying operational readiness.

4. Time Required:

   a. EO = 75 x 50 min periods; and

   b. EC = 8 x 50 min periods.

5. Instructional Methods:

   a. Lecture;

   b. Demonstration;

   c. Trainee practice; and

   d. EC.

6. Training Aids:

   a. MPT or Radar;

   b. References;

   c. Audio visual aids; and

   d. Handouts.

7. Test Details:

   a. EO 002.03 will be tested in EC 002.03 and as part of PC 002 IAW Annex H; and

   b. EO 002.03 will have a 50 min time limit (corrective measures start at minute 45).

8. References:

   a. A01 – A26;

   b. A33 – A39; and

   c. C07 – C08.

9. Remarks:

   a. MPT may be utilized;

   b. Projects and exercises may be assigned; and

   c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

   a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Explain the components in the Medium Range Radar TX / RX	D	1	
TP2	Localize fault to LRU	D P	1 74	
TP3	Proper removal and replacement of LRU	D P	0.5	
TP4	Verify operational readiness	D P	0.5 2	

PO 002 Maintain the Medium Range Radar System

EO 002.04 Repair the Medium Range Radar Signal Data Processor (SDP)

1. Performance: Repair the Medium Range Radar Signal Data Processor (SDP).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) Medium Range Radar System / MPT; and

      (7) Assistance.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The trainee shall repair the Medium Range Radar SDP, IAW references, by:

   a. Explain the components in the Medium Range Radar SDP;

   b. Localize fault to LRU;

   c. Proper removal and replacement of LRU; and

   d. Verify operational readiness.

4. Time Required:

   a. EO = 11 x 50 min periods.

5. Instructional Methods:

a. Lecture;

b. Demonstration;

c. Trainee practice; and

d. EC / PC.

6. Training Aids:

a. MPT or Radar;

b. References;

c. Audio visual aids; and

d. Handouts.

7. Test Details:

a. EO 002.04 will be tested in EC 002.04 as part of PC 002 IAW Annex H.

8. References:

a. A01 – A26; and

b. A33 – A40.

9. Remarks:

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Explain the components in the Medium Range Radar SDP.	L	2	

TP2	Localize fault to LRU.	D P	1 5	
TP3	Proper removal and replacement of LRU.	D P	0.5 1	
TP4	Verify operational readiness.	D P	0.5 1	

PO 002 Maintain the Medium Range Radar System

EO 002.05 Repair the Medium Range Radar Peripheral Equipment

1. Performance: Repair the Medium Range Radar Peripheral Equipment.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) Medium Range Radar System / MPT; and

      (7) Assistance.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The trainee shall repair the Medium Range Radar Peripheral Equipment, IAW references, by:

   a. Explaining the components in the Medium Range Radar Peripheral Equipment;

   b. Localizing fault to LRU;

   c. Removal and replacement of LRU; and

   d. Verifying operational readiness.

4. Time Required:

   a. EO = 36 x 50 min periods;

b. EC = 8 x 50 min periods; and

c. PC = 8 x 50 min periods.

5. Instructional Methods:

a. Lecture;

b. Demonstration;

c. Trainee practice; and

d. EC / PC.

6. Training Aids:

a. MPT or Radar;

b. References;

c. Audio visual aids; and

d. Handouts.

7. Test Details:

a. EO 002.05 will be tested in EC 002.05 as part of PC 002 IAW Annex H.

8. References:

a. A01 – A26;

b. A33 – A40.

9. Remarks:

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Explain the Medium Range Radar Peripheral Equipment, to include: 1. PCC; 2. AE; 3. RCU; 4. Cooling unit; and 5. Air gas dryer.	L	2	
TP2	Localize fault to LRU.	D P	1 24	
TP3	Proper removal and replacement of LRU.	D P	1 2	
TP4	Verify operational readiness.	D P	1 4	

PO 002 Maintain the Medium Range Radar System

EO 002.06 Perform Preventative Maintenance (PM) on the Medium Range Radar System

1. Performance: Perform PM on the 2D Radar System.

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) DWAN;

        (4) Tools and test equipment;

        (5) PPE;

        (6) Medium Range Radar System / MPT; and

        (7) Assistance.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The trainee shall perform PM on the Medium Range Radar System, IAW references, by:

    a. Explaining content and use of PM routines (NYs, NKs, and NTs);

    b. Describe the various types of PM routines;

    c. Explain a complex procedure utilizing the PM manual; and

    d. Demonstrate / practice a PM routine.

4. Time Required:

    a. EO = 6 x 50 min periods.

5. Instructional Methods:

   a. Lecture;

   b. Demonstration; and

   c. Trainee practice.

6. Training Aids:

   a. MPT or Radar;

   b. References;

   c. Audio visual aids; and

   d. Handouts.

7. Test Details:

   a. EO 002.06 will not be tested.

8. References:

   a. A01 – A26;

   b. A33 – A39;

   c. C07 – C08.

9. Remarks:

   a. MPT may be utilized;

   b. Projects and exercises may be assigned; and

   c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

   a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Explain content and use of PM routines (NYs, NKs, and NTs).	L	1	

TP2	Describe the various types of PM routines.	L	1	
TP3	Explain a complex procedure utilizing the PM manual	L	1	
TP4	Demonstrate / practice a PM routine	D P	1 2	

PO 003 – Maintain Identification Friend or Foe System (IFF)

EO 003.01 Describe the IFF System

1. Performance: Describe the IFF System.

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) DWAN;

        (4) Tools and test equipment;

        (5) PPE;

        (6) IFF / MPT; and

        (7) Assistance.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The trainee shall describe the IFF System, IAW references, by:

    a. Describe the IFF System, to include:

        (1) System block diagram.

    b. Capabilities and limitations;

    c. Theory of operation; and

    d. Controls and indicators.

4. Time Required:

    a. EO = 8 x 50 min periods;

b. EC = 1 x 50 min periods.

5. Instructional Methods:

a. Interactive lecture;

b. Demonstration;

c. Trainee practice; and

d. EC / PC.

6. Training Aids:

a. MPT or IFF System;

b. References;

c. Audio visual aids; and

d. Handouts.

7. Test Details:

a. EO 003.01 will be tested in EC 003.01 and as part of PC 003 IAW Annex H.

8. References:

a. A01 – A26; and

b. A40 – A45.

9. Remarks:

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Describe the IFF System, to include: 1. System block diagram	L	2	
TP2	Capabilities and limitations.	L	2	
TP3	Theory of Operation.	L	2	
TP4	Controls and indicators.	L	2	

PO 003 Maintain the Identification Friend or Foe System (IFF)

EO 003.02 Initialize the IFF System

1. Performance: Initialize the IFF System

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) DWAN;

        (4) Tools and test equipment;

        (5) PPE;

        (6) IFF / MPT; and

        (7) Assistance.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The trainee shall initialize the IFF System, IAW references, by:

    a. Explain Flash up / Shut down process;

    b. Load / verify the software; and

    c. Carry out system functional checks and verify operational readiness

4. Time Required:

    a. EO = 13 x 50 min periods; and

    b. EC = 1 x 50 min periods.

5. Instructional Methods:

a. Interactive lecture;

b. Demonstration;

c. Trainee practice; and

d. EC / PC.

6. Training Aids:

a. MPT or IFF Radar System;

b. References;

c. Audio visual aids; and

d. Handouts.

7. Test Details:

a. EO 003.02 will be tested in EC 003.02 as part of PC 003 IAW Annex H; and

b. EC 003.02 will have a 50 min time limit (corrective measures start at minute 45).

8. References:

a. A01 – A26; and

b. A40 – A45.

9. Remarks:

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Explain flash-up / shutdown process, to include: 1. Pre-operational checks; 2. System flash-up procedure; 3. System shut down procedure; and 4. Checklists.	L D P	1 1 2	
TP2	Load / verify software. 1. Cold / warm procedure; 2. Transponder control; 3. Subsystem status monitor; and 4. Operational data recording.	L P	1 4	
TP3	Carry out system functional checks and verify operational readiness.	L P	1 3	

PO 003 Maintain Identification Friend or Foe System (IFF)

EO 003.03 Repair IFF Transponder Assembly

1. Performance: Repair the IFF transponder assembly.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) IFF / MPT; and

      (7) Assistance.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The trainee shall repair the IFF transponder assembly, IAW references, by:

   a. Explain the components in the IFF transponder assembly;

   b. Localizing fault to LRU;

   c. Proper removal and replacement of LRU; and

   d. Verifying operational readiness.

4. Time Required:

   a. EO = 33 x 50 min periods; and

   b. EC = 1 x 50 min periods.

5. Instructional Methods:

a. Lecture;

b. Demonstration;

c. Trainee practice; and

d. EC / PC.

6. Training Aids:

a. MPT or IFF Radar System;

b. References;

c. Audio visual aids; and

d. Handouts.

7. Test Details:

a. EO 003.03 will be tested as part of PC 003 IAW Annex H.

8. References:

a. A01 – A26; and

b. A40 – A45.

9. Remarks:

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Explain the components of the transponder, to include: 1. Transponder (A4);	L	2	

	2. Mission / IFF Subsystem Computer (A9); 3. Control Panel (A10); and 4. Low Voltage Power Supply (A5).			
TP2	Localize fault to LRU	D P	1 6	
TP3	Proper removal and replacement of LRU	D P	0.5 2	
TP4	Verify operational readiness	D P	0.5 1	

PO 003 Maintain Identification Friend or Foe System (IFF)

EO 003.04 Repair the IFF Interrogator

1. Performance: Repair the IFF Interrogator.

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) DWAN;

        (4) Tools and test equipment;

        (5) PPE;

        (6) IFF / MPT; and

        (7) Assistance.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The trainee shall repair the IFF Interrogator, IAW references, by:

    a. Explain the components in the IFF Interrogator;

    b. Localize fault to LRU;

    c. Proper removal and replacement of LRU; and

    d. Verify operational readiness.

4. Time Required:

    a. EO = 22 x 50 min periods;

    b. EC = 8 x 50 min periods; and

c. PC = 12 x 50 min periods.

5. Instructional Methods:

a. Lecture;

b. Demonstration;

c. Trainee practice; and

d. EC / PC.

6. Training Aids:

a. MPT or IFF Radar System;

b. References;

c. Audio visual aids; and

d. Handouts.

7. Test Details:

a. EO 003.04 will be tested in EC 003.04 as part of PC 003 IAW Annex H;

b. EC 003.04 will have a 50 min time limit (corrective measures start at min 45); and

c. PC 003 will have a 50 min time limit (corrective measures start at min 45).

8. References:

a. A01 – A26;

b. A41 – A46.

9. Remarks:

a. MPT may be utilized;

b. Projects and exercises may be assigned;

c. Assignments are to be completed during non-instructional time; and

d. 8 periods for practice of overall system faults prior to PC 003.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Explain the components in the IFF Interrogator.	L	2	
TP2	Localize fault to LRU.	D P	1 14	
TP3	Proper removal and replacement of LRU.	D P	0.5 2	
TP4	Verify operational status	D P	0.5 2	

PO 003 Maintain the Medium Range Radar System

EO 003.05 Perform Preventative Maintenance (PM) on the IFF System

1. Performance: Perform PM on the IFF Interrogator.

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) DWAN;

        (4) Tools and test equipment;

        (5) PPE;

        (6) IFF / MPT; and

        (7) Assistance.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The trainee shall perform PM on IFF Interrogator, IAW references, by:

    a. Explaining content and use of PM routines (NYs, NKs, and NTs);

    b. Describe the various types of PM routines;

    c. Explain a complex procedure utilizing the PM manual; and

    d. Demonstrate / practice a PM routine.

4. Time Required:

    a. EO = 13 x 50 min periods.

5. Instructional Methods:

a. Lecture;

b. Demonstration;

c. Trainee practice; and

d. EC / PC.

6. Training Aids:

a. MPT or IFF Radar System;

b. References;

c. Audio visual aids; and

d. Handouts.

7. Test Details:

a. Nil.

8. References:

a. A01 – A26; and

b. A40 – A45.

9. Remarks:

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Explain content and use of PM routines (Nys, NKs, and NTs).	L	2	

TP2	Describe the various types of PM routines.	L	2	
TP3	Explain a complex procedure utilizing the PM manual	L	2	
TP4	Conduct a PM routine	D P	2 5	

PO 004 – Maintain the Electronic Support Measures System

EO 004.01 Describe the Electronic Support Measures System

1. Performance: Describe the Electronic Support Measures (ESM) System.

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) DWAN;

        (4) Tools and test equipment;

        (5) PPE;

        (6) ESM System / MPT; and

        (7) Assistance.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The trainee shall describe the ESM System, IAW references, by:

    a. Describe the ESM System, to include:

        (1) System block diagram.

    b. Capabilities and limitations;

    c. Theory of operation; and

    d. Controls and indicators.

4. Time Required:

    a. EO = 7 x 50 min periods.

5. Instructional Methods:

   a. Interactive lecture;

   b. Demonstration;

   c. Trainee practice; and

   d. EC / PC.

6. Training Aids:

   a. MPT or ESM System;

   b. References;

   c. Audio visual aids; and

   d. Handouts.

7. Test Details:

   a. EO 004.01 will be tested as part of PC 004 IAW Annex H.

8. References:

   a. A01 – A26; and

   b. A46 – A49.

9. Remarks:

   a. MPT may be utilized;

   b. Projects and exercises may be assigned; and

   c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

   a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Describe the ESM System, to include: 1. System block diagram	L	3	

TP2	Capabilities and limitations, to include: 1. Operator workstation; and 2. System specifications.	L	1	
TP3	Theory of Operation.	L	1	
TP4	Controls and indicators.	L	2	

PO 004 Maintain the Electronic Support Measures System

EO 004.02 Initialize the ESM System

1. Performance: Initialize the ESM System

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) DWAN;

        (4) Tools and test equipment;

        (5) PPE;

        (6) ESM System / MPT; and

        (7) Assistance.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The trainee shall initialize the ESM System, IAW references, by:

    a. Explain Flash up / Shut down process;

    b. Load / verify the software; and

    c. Carry out system functional checks and verify operational readiness

4. Time Required:

    a. EO = 4 x 50 min periods.

5. Instructional Methods:

    a. Interactive lecture;

b. Demonstration;

c. Trainee practice; and

d. EC / PC.

6. Training Aids:

a. MPT or ESM System;

b. References;

c. Audio visual aids; and

d. Handouts.

7. Test Details:

a. EO 004.02 will be tested as part of PC 004 IAW Annex H.

8. References:

a. A01 – A26; and

b. A46 – A49.

9. Remarks:

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Explain flash-up / shutdown process.	L	1	
TP2	Load / verify software.	L	0.5	

TP3	Carry out system functional checks and verify operational readiness.	L D P	0.5 1 1	

PO 004 Maintain Electronic Support Measures System

EO 004.03 Repair the ESM Reception Group

1. Performance: Repair the ESM Reception Group Function.

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) DWAN;

        (4) Tools and test equipment;

        (5) PPE;

        (6) ESM System / MPT; and

        (7) Assistance.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The trainee shall repair the ESM Reception Group function, IAW references, by:

    a. Explain the components of the ESM Reception Group, to include:

        (1) Omni-directional Antenna Unit (OAU); and

        (2) Direction Finding Antenna (DFA).

    b. Localizing fault to LRU;

    c. Proper removal and replacement of LRU; and

    d. Verifying operational readiness.

4. Time Required:

a. EO = 33 x 50 min periods.

5. Instructional Methods:

a. Lecture;

b. Demonstration;

c. Trainee practice; and

d. EC / PC.

6. Training Aids:

a. MPT or ESM System;

b. References;

c. Audio visual aids; and

d. Handouts.

7. Test Details:

a. EO 004.03 will be tested as part of PC 003 IAW Annex H.

8. References:

a. A01 – A26; and

b. A46 – A49.

9. Remarks:

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Explain the components of the ESM Reception Group, to include: 1. Omni-directional antenna unit (OAU), to include: a. OAU bottom connector unit; and b. OAU FER Components. 2. Directional Finding Antenna (DFA), to include: a. HB antenna; b. LB antenna; c. MB antenna; d. VHB antenna; and e. RVU to Video Unit, to include: (1) RFU	L	10	
TP2	Localize fault to LRU, to include: 1. symptom recognition; 2. use of fault finding logic charts; 3. convenient and appropriate test points; 4. logical procedure; and 5. Interpretation of Functional Block Diagrams.	D P	3 12	
TP3	Proper removal and replacement of LRU	D	1	

TP4	Verify operational readiness	D P	1 6	

PO 004 Maintain the Electronic Support Measures System

EO 004.04 Repair the ESM Signal Measurement Group

1. Performance: Repair the ESM signal measurement group.

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) DWAN;

        (4) Tools and test equipment;

        (5) PPE;

        (6) ESM with IFM / PME faults / MPT; and

        (7) Assistance.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The trainee shall repair the ESM signal measurement group, IAW references, by:

    a. Explain the components of the ESM signal measurement group, to include:

        (1) RDU; and

        (2) DRPU.

    b. Localize fault to LRU, to include:

        (1) RDU;

        (2) RCB; and

(3) DRPU.

c. Proper removal and replacement of LRU; and

d. Verify operational readiness, to include:

(1) WBR Path DF Channels Calibration;

(2) NBR Path DF Channels Calibration (MB, HB); and

(3) Amplitude Calibration.

4. Time Required:

a. EO = 30 x 50 min periods; and

b. EC = 8 x 50 min periods.

5. Instructional Methods:

a. Lecture;

b. Demonstration;

c. Trainee practice; and

d. EC / PC.

6. Training Aids:

a. MPT or ESM System;

b. References;

c. Audio visual aids; and

d. Handouts.

7. Test Details:

a. EO 004.04 will be tested in EC 004.04 as part of PC 004 IAW Annex H; and

b. EC 003.04 will have a 50 min time limit (corrective measures start at min 45).

8. References:

a. A01 – A26; and

b. A46.

9. Remarks:

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Explain the components of the ESM Signal Measurement Group, to include: 1. RDU; and 2. DRPU.	L	7	
TP2	Localize fault to LRU, to include: 1. RDU; 2. RCB; and 3. DRPU.	L D P	1 1 12	
TP3	Proper removal and replacement of LRU.	D P	0.5 2	
TP4	Verify operational readiness, to include: 1. WBR Patch DF Channels Calibration; 2. NBR Path DF Channels Calibration (MB, HB); and 3. Amplitude Calibration.	L D P	1 0.5 5	

PO 004 Maintain the Electronic Support Measures System

EO 004.05 Repair the ESM Signal Processing Group

1. Performance: Repair the ESM Signal Processing Group.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) ESM System / MPT; and

      (7) Assistance.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The trainee shall perform repair the ESM Signal Processing Group, IAW references, by:

   a. Explaining the components of the ESM Signal Processing Group, to include:

      (1) ABJB;

      (2) Maintenance Panel;

      (3) Fan units;

      (4) Power supply;

      (5) ISU;

(6) PCU; and

(7) UPS.

b. Localizing fault to LRU, to include:

(1) ISU;

(2) ESU; and

(3) PSU.

c. Proper removal and replacement of LRU; and

d. Verify operational readiness.

4. Time Required:

a. EO = 20 x 50 min periods; and

b. PC = 8 x 50 min periods.

5. Instructional Methods:

a. Lecture;

b. Demonstration;

c. Trainee practice; and

d. EC / PC.

6. Training Aids:

a. MPT or ESM System;

b. References;

c. Audio visual aids; and

d. Handouts.

7. Test Details:

a. EO 004.05 will be tested in EC 004.05 as part of PC 004 IAW Annex H; and

b. PC 004 will have a 50 minute time limit (corrective measures start at min 45).

8. References:

a. A01 – A26; and

b. A46.

9. Remarks:

a. MPT may be utilized;

b. Projects and exercises may be assigned;

c. Assignments are to be completed during non-instructional time;

d. 8 periods to practice overall system faults.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Explain the components of Signal Processing Group to include: 1. ABJB; 2. maintenance panel; 3. fan units; 4. power supply; 5. ISU; 6. PCU; and 7. UPS.	L	3	
TP2	Localize fault to LRU, to include: 1. ISU; 2. ESU; and 3. PSU.	D P	1 13	

TP3	Proper removal and replacement of LRU	D P	0.5 1	
TP4	Verify operational readiness.	D P	0.5 1	

PO 004 – Maintain the Electronic Support Measures System

EO 004.06 Perform Preventative Maintenance on the ESM

1. Performance: Perform PM on the ESM.

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) DWAN;

        (4) Tools and test equipment;

        (5) PPE;

        (6) ESM System / MPT; and

        (7) Assistance.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The trainee shall perform PM on ESM, IAW references, by:

    a. Explaining content and use of PM routines (NYs, NKs, and NTs);

    b. Describe the various types of PM routines;

    c. Explain a complex procedure utilizing the PM manual; and

    d. Demonstrate / practice a PM routine.

4. Time Required:

    a. EO = 13 x 50 min periods.

5. Instructional Methods:

a. Lecture;

b. Demonstration;

c. Trainee practice; and

d. EC / PC.

6. Training Aids:

a. MPT or ESM System;

b. References;

c. Audio visual aids; and

d. Handouts.

7. Test Details:

a. Nil.

8. References:

a. A01 – A26; and

b. A46.

9. Remarks:

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Explain content and use of PM routines (NYs, NKs, and NTs).	L	2	

TP2	Describe the various types of PM routines.	L	2	
TP3	Explain a complex procedure utilizing the PM manual	L	2	
TP4	Conduct a PM routine	D P	2 5	

PO 005 – Maintain the Electronic Support Measures System (Sea Search)

EO 005.01 Describe the Electronic Support Measures System (Sea Search)

1. Performance: Describe the ESM System (Sea Search).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Replacement parts;

      (4) Tools and test equipment;

      (5) software;

      (6) AN/ULR 501 and/or MPT; and

      (7) Assistance.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The trainee shall describe the ESM (Sea Search) System, IAW references, by:

   a. Describe the ESM (Sea Search) System, to include:

      (1) Publication layout;

      (2) System block diagram;

      (3) Capabilities and limitations;

      (4) Operations;

      (5) Antenna;

      (6) RFD;

(7) Tuner;

(8) Data processor; and

(9) Remote terminal.

4. Time Required:

a. EO = 10 x 50 min periods.

5. Instructional Methods:

a. Interactive lecture.

6. Training Aids:

a. MPT and/or ESM System AN/ULR 501 Sea Search;

b. References;

c. Audio visual aids; and

d. Handouts.

7. Test Details:

a. EO 005.01 will be tested in EC 005.01 as part of PC 005 IAW Annex H.

8. References:

a. A01 – A26; and

b. C01 – C06.

9. Remarks:

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Describe the ESM System, to include: 1. Publication layout; 2. System block diagram; 3. Capabilities and limitations; 4. Operation; 5. Antenna; 6. RFD; 7. Tuner; 8. Data processor; and 9. Remote terminal diagnostics.	L	10	

PO 005 Maintain the Electronic Support Measures System (Sea Search)

EO 005.02 Initialize the ESM System AN/ULR 501 (Sea Search)

1. Performance: Initialize the ESM System An/ULR 501.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Replacement parts;

      (4) Software;

      (5) Tools and test equipment;

      (6) AN/ULR 501 and/or MPT; and

      (7) Assistance.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The trainee shall initialize the AN/ULR 501, IAW references, by:

   a. Powering up;

   b. Checking during and after power up;

   c. Loading and verifying system software;

   d. Conducting system functional checks;

   e. Reporting equipment status to supervisor; and

   f. Powering down.

4. Time Required:

a. EO = 6 x 50 min periods.

5. Instructional Methods:

a. Demonstration; and

b. Trainee practice.

6. Training Aids:

a. MPT and/or ESM System AN/ULR 501 Sea Search;

b. References;

c. Audio visual aids; and

d. Handouts.

7. Test Details:

a. EO 005.02 will be tested as part of PC 005 IAW Annex H.

8. References:

a. A01 – A26; and

b. C01 – C06.

9. Remarks:

a. Practice flash-up and faultfinding will be conducted in teams of 2;

b. EC / PCs will be carried out individually; and

c. Instructor must ensure system is initialized to reduce training delay.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Initialize ESM (Sea Search), to include: 1. Power-up;	D P	2 4	

	2. Checks during and after power-up; 3. Load and verify system software; 4. Conduct system functional checks; 5. Report System Status to supervisor; and 6. Power down.			

PO 005 Maintain Electronic Support Measures System (Sea Seach)

EO 005.03 Repair the ESM System (Sea Search)

1. Performance: Repair the ESM System (Sea Search).

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) Software;

        (4) Tools and test equipment;

        (5) Replacement parts;

        (6) AN/ULR 501 and/or MPT; and

        (7) Assistance.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The trainee shall maintain the AN/ULR 501, IAW references, by:

    a. Localizing fault to LRU;

    b. Proper removal and replacement of LRU; and

    c. Verifying operational readiness.

4. Time Required:

    a. EO = 28 x 50 min periods; and

    b. PC = 8 x 50 min periods.

5. Instructional Methods:

a. Lecture;

b. Demonstration;

c. Trainee practice; and

d. PC.

6. Training Aids:

a. MPT and/or ESM System AN/ULR 501;

b. References;

c. Audio visual aids; and

d. Handouts.

7. Test Details:

a. EO 005.03 will be tested as part of PC 003 IAW Annex H; and

b. PC 005 and debrief to e conducted IAW Annex H with a 50 min time limit (corrective measures start at min 45).

8. References:

a. A01 – A26; and

b. C01 – C06.

9. Remarks:

a. Practice flash-up and faultfinding will be conducted in teams of 2;

b. EC/PCs will be carried out individually; and

c. 12 periods are allocated to practice faults. 4 periods are allocated for overall system faults.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Localize fault to LRU, by: 1. Symptom recognition; 2. Use of fault-finding logic charts; 3. Convenient and appropriate test points; 4. Logical procedure; 5. Interpretation of schematics; 6. Verify operational readiness; 7. Describe tests, trials, and alignments; and 8. Describe removal and replacement of LRUs.	D P	1 24	
TP2	Proper removal and replacement of LRU.	D P	0.5 1	
TP3	Verify operational readiness	D P	0.5 1	

PO 005 Maintain the Electronic Support Measures System (Sea Seach)

EO 005.04 Perform PM on the ESM (Sea Seach)

1. Performance: Perform PM o the ESM (Sea Search).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) ESM AN/ULR 501 and/or MPT; and

      (7) Assistance.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The trainee shall perform PM on the ESM (Sea Search), IAW references, by utilizing:

   a. Explain content and use of PM routines (NYs, NKs and NTs);

   b. describe the various types of PM routines;

   c. explain a complex procedure utilizing the PM manual; and

   d. demonstrate/practice a PM routine.

4. Time Required:

   a. EO = 3 x 50 min periods.

5. Instructional Methods:

   a. Lecture;

   b. Demonstration;

   c. Trainee practice; and

   d. EC / PC.

6. Training Aids:

   a. MPT or ESM (Sea Search) System;

   b. References;

   c. Audio visual aids; and

   d. Handouts.

7. Test Details:

   a. Nil.

8. References:

   a. A01 – A26; and

   b. C01 – C06.

9. Remarks:

   a. MPT may be utilized;

   b. Projects and exercises may be assigned; and

   c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

   a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Explain content and use of PM routines (NYs, NKs, and NTs).	L	0.5	

TP2	Describe the various types of PM routines.	L	0.5	
TP3	Explain a complex procedure utilizing the PM manual	L	0.5	
TP4	Conduct a PM routine	D P	0.5 1	

PO 006 – Maintain Communications Intercept System

EO 006.01 Describe the Communications Intercept System (CIS)

1. Performance: Describe the operation of the Communications Intercept System.

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) Replacement parts;

        (4) Tools and test equipment;

        (5) software;

        (6) CIS and/or MPT; and

        (7) Assistance.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The trainee shall describe the operation of the CIS, IAW references, by:

    a. System overview;

    b. Capabilities and limitations;

    c. Theory of operations; and

    d. Controls and indicators.

4. Time Required:

    a. EO = 7 x 50 min periods.

5. Instructional Methods:

   a. Interactive lecture.

6. Training Aids:

   a. MPT and/or Communications Intercept System;

   b. References;

   c. Audio visual aids; and

   d. Handouts.

7. Test Details:

   a. EO 006.01 will be tested as part of PC 006.

8. References:

   a. A01 – A26; and

   b. A51 – A56.

9. Remarks:

   a. Practice Flash-up and faultfinding will be conducted in teams of 2; and

   b. EC / PCs will be carried out individually.

10. Teaching Points:

    a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	System overview, to include: 1. Major components; 2. Locations on the ship; and 3. Direction finder set.	L	7	
TP2	Capabilities and limitations	L	2	

TP3	Theory of operation	L	2	
TP4	Controls and indicators	L	2	

PO 006 Maintain the Communications Intercept System

EO 006.02 Initialize the Communications Intercept System

1. Performance: Initialize the CIS System AN/SRD 504.

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) Replacement parts;

        (4) Software;

        (5) Tools and test equipment;

        (6) AN/SRD 504 and/or MPT; and

        (7) Assistance.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The trainee shall initialize the CIS, IAW references, by:

    a. Explaining flash up/Shut down process; and

    b. Demonstrating and practicing Flash up / Shut down.

4. Time Required:

    a. EO = 7 x 50 min periods.

5. Instructional Methods:

    a. Demonstration; and

    b. Trainee practice.

6. Training Aids:

   a. MPT and/or CIS;

   b. References;

   c. Audio visual aids; and

   d. Handouts.

7. Test Details:

   a. EO 006.02 will be tested as part of PC 006 IAW Annex H.

8. References:

   a. A01 – A26; and

   b. A51 – A56.

9. Remarks:

   a. Practice flash-up and faultfinding will be conducted in teams of 2; and

   b. EC / PCs will be carried out individually.

10. Teaching Points:

   a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Explain the Flashup / shutdown process, to include: 1. System AC power up; 2. DF System initialization; 3. DF operations; 4. Acquisition operations; 5. Mapping; 6. System AC shut down.	L	1	

TP2	Demonstrate and practice flash up / shutdown	D P	1 5	

PO 006 Maintain Communications Intercept System

EO 006.03 Repair the Communications Intercept System

1. Performance: Repair the Communications Intercept System AN/SRD 504.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Software;

      (4) Tools and test equipment;

      (5) Replacement parts;

      (6) CIS and/or MPT; and

      (7) Assistance.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The trainee shall repair the CIS, IAW references, by:

   a. Localizing fault to LRU, to include:

      (1) Utilizing fault finding flowcharts;

      (2) Check visual indicators;

      (3) Observe BIT results; and

      (4) Interpret diagrams.

   b. Proper removal and replacement of LRU; and

   c. Verifying operational readiness.

4. Time Required:

   a. EO = 30 x 50 min periods; and

   b. PC = 8 x 50 min periods.

5. Instructional Methods:

   a. Lecture;

   b. Demonstration;

   c. Trainee practice; and

   d. PC.

6. Training Aids:

   a. MPT and/or ESM CIS;

   b. References;

   c. Audio visual aids; and

   d. Handouts.

7. Test Details:

   a. PC 006 will be tested IAW Annex H.

8. References:

   a. A01 – A26; and

   b. A51 – A56.

9. Remarks:

   a. Practice flash-up and faultfinding will be conducted in teams of 2;

   b. EC/PCs will be carried out individually; and

   c. 16 periods are allocated to practice faults. 4 periods are allocated for overall system faults.

10. Teaching Points:

   a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Localize fault to LRU, by: 1. Utilizing fault finding flowchart; 2. Check visual indications; 3. Observe BITE results; and 4. Interpret diagrams.	L D	4 6	
TP2	Proper removal and replacement of LRU.	P	20	
TP3	Verify operational readiness			

PO 006 Maintain the Communications Intercept System

EO 006.04 Perform PM on the Communications Intercept System

1. Performance: Perform PM on the CIS.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) CIS and/or MPT; and

      (7) Assistance.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The trainee shall perform PM on the CIS, IAW references, by utilizing:

   a. Explain content and use of PM routines (NYs, NKs and NTs);

   b. describe the various types of PM routines;

   c. explain a complex procedure utilizing the PM manual; and

   d. demonstrate/practice a PM routine.

4. Time Required:

   a. EO = 2 x 50 min periods.

5. Instructional Methods:

   a. Lecture;

   b. Demonstration;

   c. Trainee practice; and

   d. EC / PC.

6. Training Aids:

   a. MPT or CIS;

   b. References;

   c. Audio visual aids; and

   d. Handouts.

7. Test Details:

   a. Nil.

8. References:

   a. A01 – A26; and

   b. A51 – A56.

9. Remarks:

   a. MPT may be utilized;

   b. Projects and exercises may be assigned; and

   c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

   a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Explain content and use of PM routines (NYs, NKs, and NTs).	L	2	

TP2	Describe the various types of PM routines.	L	2	
TP3	Explain a complex procedure utilizing the PM manual	L	2	
TP4	Conduct a PM routine	D P	2 5	

PO 007 Maintain the Combat Management System

EO 007.01 Describe the Combat Management System

1. Performance: Describe the Combat Management System (CMS).

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) Replacement parts;

        (4) Tools and test equipment;

        (5) software;

        (6) MPT; and

        (7) Assistance.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The trainee shall describe the CMS, IAW references, by:

    a. System overview;

    b. Capabilities and limitations;

    c. Theory of operations;

    d. Power distribution; and

    e. Controls and indicators.

4. Time Required:

    a. EO = 10 x 50 min periods.

5. Instructional Methods:

   a. Interactive lecture.

6. Training Aids:

   a. MPT and/or CMS;

   b. References;

   c. Audio visual aids; and

   d. Handouts.

7. Test Details:

   a. EO 007.01 will be tested as part of PC 007.

8. References:

   a. A01 – A26;

   b. A57 – A66; and

   c. C07 – C08.

9. Remarks:

   a. For the purpose of this EO, the student is assumed to have a working knowledge of CMS operation, initialization, and maintenance. This training should be focused on the higher function of CMS.

10. Teaching Points:

   a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	System overview, to include: 1. MCE components to the LRU; and 2. MFW components to the LRU.			
TP2	Capabilities and limitations, to include:	L	1	

	1. Cabinet enclosures; 2. Remote access server; 3. Cabinet cooling system; and 4. MCE cabinets.			
TP3	Theory of operation, to include: 1. Layered system of sensors; 2. Information exchange system (IES); 3. Tracking tools; and 4. Decision aids and weapons, to include: a. Detect; b. Locate; c. Track; d. Identify; e. Assess; and f. Engage threats.			
TP4	Power distribution, to include: 1. Breaker and switchboard connection; 2. MCE power; 3. MFW power; and 4. Subassembly voltage input.			
TP5	Controls and indicators, to include: 1. Common MCE Components; and 2. MCE subassemblies.			

PO 007 Maintain Combat Management System

EO 007.02 Initialize the Combat Management System

1. Performance: Initialize the CMS.

2. Conditions:

    a. Given:

        (1) Supervision;

        (2) References;

        (3) Replacement parts;

        (4) Software;

        (5) Tools and test equipment;

        (6) CMS and/or MPT; and

        (7) Assistance.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Any shipboard conditions at sea or alongside; and

        (2) Day or night.

3. Standard: The trainee shall initialize the CMS, IAW references, by:

    a. Initialize CMS, to include:

        (1) from cold boot;

        (2) from warm boot/software reload;

        (3) shutdown; and

        (4) shutdown into operational load/unload.

4. Time Required:

    a. EO = 10 x 50 min periods.

5. Instructional Methods:

a. Demonstration; and

b. Lecture.

6. Training Aids:

a. MPT and/or CMS;

b. References;

c. Audio visual aids; and

d. Handouts.

7. Test Details:

a. EO 007.02 will be tested as part of PC 007 IAW Annex H.

8. References:

a. A01 – A26;

b. A57 – A66; and

c. C07 – C08.

9. Remarks:

a. EC / PCs will be carried out individually.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Initialize CMS, to include: 1. From cold boot; 2. From warm boot software reload; 3. Shutdown; and 4. Shutdown into operational load / unload.	D P	1 9	

PO 007 Maintain the Combat Management System

EO 007.03 Repair the Combat Management System

1. Performance: Repair the CMS.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Software;

      (4) Tools and test equipment;

      (5) Replacement parts;

      (6) CMS and/or MPT; and

      (7) Assistance.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The trainee shall repair the CMS, IAW references, by:

   a. Repairing MCEs, to include:

      (1) System recognition;

      (2) Localizing fault to LRU;

      (3) Use fault finding logic charts;

      (4) Logical procedure;

      (5) Interpretation of schematics;

      (6) Verify operational readiness;

(7) Tests and alignments;

(8) Describe removal and replacement of LRU; and

(9) Verify operational readiness.

b. repairing MFWs, to include same as MCE;

c. repairing peripherals, to include same as MCE; and

d. repairing Network switches and firewalls, to include same as MCE.

4. Time Required:

a. EO = 12 x 50 min periods; and

b. PC = 5 x 50 min periods.

5. Instructional Methods:

a. Lecture;

b. Demonstration; and

c. PC.

6. Training Aids:

a. MPT and/or CMS;

b. References;

c. Audio visual aids; and

d. Handouts.

7. Test Details:

a. EO 007.03 will be tested with PC 007 IAW Annex H.

8. References:

a. A01 – A26;

b. A57 – A66; and

c. C07 – C08.

9. Remarks:

a. EC/PCs will be carried out individually.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Repair MCEs, by: 1. System recognition; 2. Localizing fault to LRU; 3. Use fault finding logic charts; 4. Logical procedure; 5. Interpretation of schematics; 6. Verify operational readiness; 7. Tests and alignments; 8. Describe removal and replacement of LRU; 9. Verify operational readiness.	P	3	
TP2	Repair MFWs, by: 1. System recognition; 2. Localizing fault to LRU; 3. Use fault finding logic charts; 4. Logical procedure; 5. Interpretation of schematics; 6. Verify operational readiness; 7. Tests and alignments; 8. Describe removal and replacement of LRU;	P	3	

	9. Verify operational readiness.			
TP3	Repair peripherals (CID / LSD), by: 1. System recognition; 2. Localizing fault to LRU; 3. Use fault finding logic charts; 4. Logical procedure; 5. Interpretation of schematics; 6. Verify operational readiness; 7. Tests and alignments; 8. Describe removal and replacement of LRU; 9. Verify operational readiness.	P	3	
TP4	Repair Network, switches, and firewalls, by: 1. System recognition; 2. Localizing fault to LRU; 3. Use fault finding logic charts; 4. Logical procedure; 5. Interpretation of schematics; 6. Verify operational readiness; 7. Tests and alignments; 8. Describe removal and replacement of LRU; 9. Verify operational readiness	P	3	

PO 007 Maintain the Combat Management System

EO 007.04 Perform PM on the Combat Management System

1. Performance: Perform PM on the CIS.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) DWAN;

      (4) Tools and test equipment;

      (5) PPE;

      (6) CMS and/or MPT; and

      (7) Assistance.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The trainee shall perform PM on the CMS, IAW references, by utilizing:

   a. Explain content and use of PM routines (NYs, NKs and NTs);

   b. describe the various types of PM routines;

   c. explain a complex procedure utilizing the PM manual; and

   d. demonstrate/practice a PM routine.

4. Time Required:

   a. EO = 3 x 50 min periods.

5. Instructional Methods:

   a. Lecture;

   b. Demonstration;

   c. Trainee practice; and

   d. EC / PC.

6. Training Aids:

   a. MPT or CMS;

   b. References;

   c. Audio visual aids; and

   d. Handouts.

7. Test Details:

   a. Nil.

8. References:

   a. A01 – A26;

   b. A57 – A66; and

   c. C07 – C08.

9. Remarks:

   a. MPT may be utilized;

   b. Projects and exercises may be assigned; and

   c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

   a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	Explain content and use of PM routines (NYs, NKs, and NTs).	L	0.5	
TP2	Describe the various types of PM routines.	L	0.5	
TP3	Explain a complex procedure utilizing the PM manual	L	0.5	
TP4	Conduct a PM routine	D P	0.5 1	

PO 008 – Utilize DRMIS Portal as a Maintenance Supervisor IAW QSP 119097

EO 008.01 Utilize DRMIS Portal as a Maintenance Supervisor IAW QSP 119097

1. Performance: Utilize DRMIS Portal as a Maintenance Supervisor IAW QSP DRMIS QSP 119097 – Eos 002.01,.02,.03.

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) Computer with access to DRMIS simulator; and

      (3) Supervision.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom.

3. Standard: Utilize DRMIS Portal as a Maintenance Supervisor IAW QSP DRMIS QSP 119097 – Eos 002.01,.02,.03.

4. Time Required:

   a. EO = 5 x 50 min periods; and

   b. PC = 1 x 50 min periods.

5. Instructional Methods:

   a. IAW EOs 002.01, 002.03, and 002.03 DRMIS QSP 119097.

6. Training Aids:

   a. IAW EOs 002.01, 002.03, and 002.03 DRMIS QSP 119097.

7. Test Details:

   a. IAW EOs 002.01, 002.03, and 002.03 DRMIS QSP 119097.

8. References:

a. IAW EOs 002.01, 002.03, and 002.03 DRMIS QSP 119097.

9. Remarks:

a. All trainees require level 1 security clearance.

10. Teaching Points:

a. IAW EOs 002.01, 002.03, and 002.03 DRMIS QSP 119097.

QUALIFICATION STANDARD ANNEXES

MARPAC RDIMS #
Chap 1 & 2 approved DD MM YYYY
Chap 3 & 4 approved DD MM YYYY

ANNEX A – MAIN REFERENCES

1. This annex lists all references used in this QS and TP.

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A		**Official CAF Publications and Directives that relate to this Qualification**
A01	C-01-000-000/AG-001	TECHNICAL PUB INDEX (1)
A02	C-01-000-000/AG-002	TECHNICAL PUB INDEX (2)
A03	C-01-000-000/AG-003	PUB CROSS REFERENCE LIST
A04	C-01-000-012/AG-000	NDID SYSTEM
A05	C-02-015-001/AG-000	UNSAT COND REPORTS (UCRs)
A06	C-03-005-012/AM-001	NAMMS MANUAL (1) POLICY
A07	C-03-005-012/AM-002	NAMMS (2) SMISS
A08	C-03-005-012/AM-007	NAMMS (7)
A09	C-03-007-153/AX-001	CANAVMOD INDEX
A10	C-02-040-003/TP-000	Radiation Hazard Safety Officer Handbook
A11	MARCORDS 66-5 Vol 1	Hazardous Material
A12	MARCORDS 66-8	Hazardous Material
A13	MARCORDS CS-1 Vol 2	
A14	MARCORDS Vol 2	
A15	A-LM-007-005/AX-001	Emergency Response Guide
A16	A-LM-181-001/JS-001	Supply Instructions
A17	A-LM-187-004/JS-001	Hazardous Material Guide Sheets
A18	A-LP-005-000/AG-001	Logistic Support
A19	A-MD-098-000/JD-000	HAZMAT Injures Handbook for Pre-Hospital Care
A20	C-01-000-000/DA-001	NDID Coding System
A21	SSO	Class Specific Ship's Standing Orders
A22	DAOD 4003-1	Hazardous Material Management
A23	CFMO 40-04	
A24	CFMO 40-07	
A25	ERT Pack	Platform Specific ERT Pack
A26	A-PD-055-002/PP-002	Canadian Forces Manual of Military Occupational Structure, Vol 2, Pt 2, Occupational Specification for the Naval Electronics Technician (Tactical) Occupation, MOC 285, 15 May '99, (Under Amendment)
Maintain the Long Range Radar System		
A27	C-59-155-000/MB-001	Operating Instructions Smart-S Mk2 3D Radar
A28	C-59-155-000/MD-001	Equipment Description Smart-S Mk2 3D Radar
A29	C-59-155-000/MM-001	First Line Corrective Maintenance Instructions Smart-S Mk2 3D Radar
A30	C-59-155-000/MY-001	Parts Identification List Smart-S Mk2 3D Radar

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A30	C-59-155-000/NK-001	Naval Preventive Maintenance Performance Test (Tactical) Smart-S Mk2 3D Radar
A32	C-59-155-000/NY-Z01	Naval Preventive Maintenance Schedules Smart-S Mk2 3D Radar
Maintain the Medium Range Radar System		
A33	C-59-165-000/MB-001	Operating Instructions - Sea Giraffe SG-180 (H) 2D Radar System
A34	C-59-165-000/MD-001	Equipment Description - Sea Giraffe SG-180 (H) 2D Radar System
A35	C-59-165-000/MM-00	First Line Corrective Maintenance Instructions - Sea Giraffe SG-180 (H) 2D Radar
A36	C-59-165-000/MY-001	Parts Identification List - Sea Giraffe SG-180 (H) 2D Radar
A37	C-59-165-000/NK-001	Naval Preventive Maintenance Performance Test - Sea Giraffe SG-180 (H) 2D Radar
A38	C-59-165-000/NY-001	Naval Preventive Maintenance Schedule - Sea Giraffe SG-180 (H) 2D Radar
A39	C-59-165-000/NY-Z01	Naval Preventive Maintenance Schedule - Sea Giraffe SG- 180 (H) 2D Radar
Maintain Identification Friend or Foe System		
A40	C-59-179-000/MB-001	Operating Instructions - IFF Mode S5 Identification Friend Or Foe
A41	C-59-179-000/MD-001	Equipment Description - IFF Mode S5 Identification Friend Or Foe
A42	C-59-179-000/MM-001	First Line Corrective Maintenance Instructions - IFF Mode S5 Identification Friend Or Foe
A43	C-59-179-000/MY-001	Parts Identification List - IFF Mode S5 Identification Friend Or Foe
A44	C-59-179-000/NK-001	Naval Preventive Maintenance Performance Test IFF Mode S5 Identification Friend Or Foe
A45	C-59-179-000/NY-001	Naval Preventive Maintenance Schedule - IFF Mode S5 Identification Friend Or Foe
Maintain Electronic Support Measures System		
A46	C-58-648-000/MB-001	Operating Instructions Electronic Support Measures Suite
A47	C-58-648-000/MD-001	Equipment Description Electronic Support Measures Suite
A48	C-58-648-000/MM-001	First Line Corrective Maintenance Instructions AN/SLR-505 Electronic Support Measures Suite
A49	C-58-648-000/NY-001	Naval Preventive Maintenance Schedule - Electronic Support Measures Suite
A50	C-58-648-000/MY-001	Parts Identification List - Electronic Support Measures Suite

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
Communications Intercept System		
A51	C-58-393-000/MS-001	Operation and Maintenance Instructions - Direction Finder Set AN/SRD-504
A52	C-58-393-000/NG-001 Test	Sheet Naval Systems and Equipments Direction Finder Set AN/SRD-504 Applicable to HFX Class
A53	C-58-393-000/NK-001	Naval Preventive Maintenance Performance Test (Tactical) Direction Finder Set AN/SRD-504 Applicable to Halifax Class
A54	C-58-393-000/NT-001	TRIAL Agenda For Direction Finder Set AN/SRD-504 (Alongside) Applicable To Halifax Class
A55	C-58-393-000/NT-002	Trial Agenda For Direction Finder Set AN/SRD-504 (At Sea) Applicable to Halifax Class
A56	C-58-393-000/NY-001	Naval Preventive Maintenance Schedule (Tactical) Direction Finder Set, AN/SRD-504 - Applicable to Halifax Class,
B		**(Non-CAF) Government of Canada Publications & Foreign Military (Non-DND) Publications that relate to this Qualification**
B1		
B2		
B3		
C		**Civilian Publications and Manuals that relate to this Qualification**
Sea Search ESM System AN/ULR 501		
C01	164004 AN/ULR-501	Acquisition System Operator's Manual
C02	164002 AN/ULR-501	Acquisition System Maintenance Manual
C03	164244 TN-628 B/G	RF Tuner Maintenance Manual
C04	164174 AM-5351/ULR-501	RF Distribution Maintenance Manual
C05	164154 AS-5246/ULR-501	Antenna Maintenance Manual
C06	164104 CP-5136/ULR-501	Data Processor Maintenance Manual
Maintain the Medium Range Radar System		
C07	4/LZUM1130166 Sea	Giraffe AMB H, Operators Training
C08	5/LZUM1130166	Sea Giraffe AMB H, Maintainer Training
D		**Other Publications and Manuals that relate to this Qualification**
D1		
D2		

NOTE: If additional references are identified during the TPWB this annex can be updated secretarially through NTG HQ.

ANNEX B – TASK LIST

1. The task list includes all of the tasks associated with this qualification, and the outcome of the Train / No-train decisions. If a task is designated as requiring training, the PO where it is covered is indicated. If a task is designated as not requiring training, the justification for that designation is provided.

2. The following table defines some of the criteria used to support a no-train designation in the task list. Other reasons may be used if required.

CRITERIA	DEFINITION
A	**Previously qualified.** Tasks and knowledge already possessed by the student by CAF and/or RCN courses, training and/or experience.
B	**Will receive on future courses.** Tasks and knowledge will be covered as part of other CAF and/or RCN courses and/or training.
C	**Difficulty.** Task is not difficult to learn; formal training is not required.
D	**Importance.** Task is not important; it has no significant impact on mission success.
E	**Frequency.** Task is rarely performed, no formal training is required.
F	**Lack of equipment.** Currently there is no equipment in the CAF inventory to train this task.

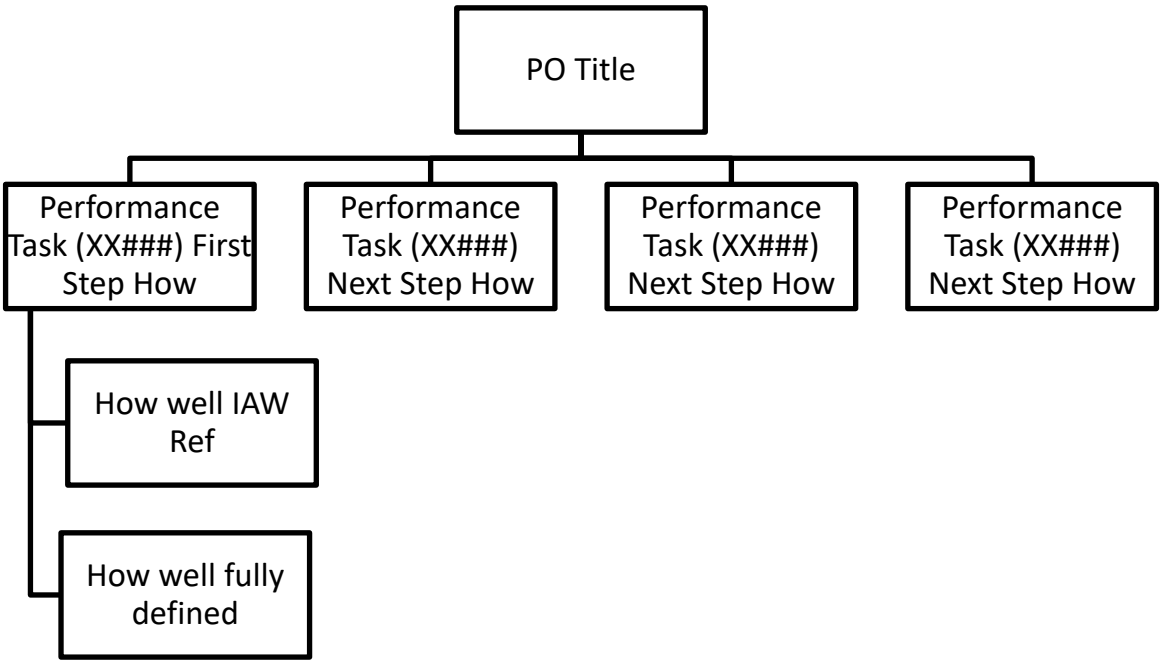
Task #	Task	PO
Duty Area D	**SMART-S Mk 2 3D Radar and 2D Radar System**	
D001	Initialize an Air Search Radar System to an operational state	
D002	Install a new or repaired/refurbished/overhauled Air Search Radar System component	
D003	Monitor an Air Search Radar System performance	
D004	Operate an Air Search Radar System	
D005	Perform post-repair calibration/testing/tuning/alignment of an Air Search Radar System	
D006	Remove a faulty component from an Air Search Radar System	
D007	Calibrate/test/tune/align a newly installed Air Search Radar System component	
D008	Conduct preliminary fault diagnostic on a Hydraulic Stabilized Platform System (Tech of the Watch)	
D009	Configure a Hydraulic Stabilized Platform System to meet an operational requirement	

D010	Configure an Air Search Radar System to meet an operational requirement	
D011	Diagnose an Air Search Radar System fault to the Lowest Repairable Unit	
D012	Initialize a Hydraulic Stabilized Platform System to an operational state	
D013	Install a new or repaired/refurbished/overhauled Hydraulic Stabilized Platform System component	
D014	Monitor a Hydraulic Stabilized Platform System performance	
D015	Perform a Hydraulic Stabilized Platform System functional check	
D016	Perform post-repair calibration/testing/tuning/alignment of a Hydraulic Stabilized Platform System	
D017	Remove a faulty component from a Hydraulic Stabilized Platform System	
D018	Calibrate/test/tune/align a newly installed Hydraulic Stabilized Platform System component	
D019	Conduct preliminary fault diagnostic on a Hydraulic Stabilized Platform System (Tech of the Watch)	
D020	Configure a Hydraulic Stabilized Platform System to meet an operational requirement	
D021	Diagnose a Hydraulic Stabilized Platform System fault to the Lowest Repairable Unit	
Duty Area O	**IFF**	
O001	Initialize an Identification Friend or Foe (IFF) System to an operational state	
O002	Install a new or repaired/refurbished/overhauled IFF System component	
O003	Monitor an IFF System performance	
O004	Operate an IFF System	
O005	Perform an IFF System functional check	
O006	Perform an IFF System PM activity (including a test/trial/inspection)	
O007	Perform post-repair calibration/testing/tuning/alignment of an IFF System	
O008	Remove a faulty component from an IFF System	
O009	Calibrate/test/tune/align a newly installed IFF System component	
O010	Conduct preliminary fault diagnostic on an IFF System (Tech of the Watch)	
O011	Configure an IFF System to meet an operational requirement	

O012	Diagnose an IFF System fault to the Lowest Repairable Unit	
Duty Area O	**CMS**	
O013	Initialize a Command and Control System (CCS) to an operational state	
O014	Monitor a CCS performance	
O015	Install a new or repaired/refurbished/overhauled CCS component	
O016	Operate a CCS	
O017	Perform a CCS functional check	
O018	Calibrate/test/tune/align a newly installed CCS component	
O019	Configure a CCS to meet an operational requirement	
O020	Diagnose a CCS fault to the Lowest Repairable Unit	
Duty Area E	**ESM**	
E001	Initialize an Electronic Support Measures (ESM) System to an operational state. (e.g. CANEWS)	
E002	Install a new or repaired/refurbished/overhauled ESM System component	
E003	Monitor an ESM System performance	
E004	Operate an ESM System	
E005	Perform an ESM System functional check	
E006	Perform an ESM System PM activity (including a test/trial/inspection)	
E007	Perform post-repair calibration/testing/tuning/alignment of an ESM System	
E008	Remove a faulty component from an ESM System	
E009	Calibrate/test/tune/align a newly installed ESM System component	
E010	Conduct preliminary fault diagnostic on an ESM System (Tech of the Watch)	
E011	Configure an ESM System to meet an operational requirement	
E012	Diagnose an ESM System fault to the Lowest Repairable Unit	
Duty Area E	**CIO**	
E013	Initialize a Direction Finder (DF) Communication Intercept Outfit (CIO) System to an operational state	
E014	Install a new or repaired/refurbished/overhauled DF (CIO) System component	
E015	Monitor a DF (CIO) System performance	
E016	Operate a DF (CIO) System	
E017	Perform a DF (CIO) System functional check	

E018	Perform a DF (CIO) System PM activity (including a test/trial/inspection)	
E019	Perform post-repair calibration/testing/tuning/alignment of a DF (CIO) System	
E020	Remove a faulty component from a DF (CIO) System	
E021	Repair/refurbish/overhaul a faulty component from a DF (CIO) System	
E022	Bench test a repaired/refurbished/overhauled DF (CIO) System component	
E023	Analyze/evaluate/interpret a DF (CIO) System functional check result	
E024	Analyze/evaluate/interpret a DF (CIO) System test/trial/inspection result or report	
E025	Calibrate/test/tune/align a newly installed DF (CIO) System component	
E026	Conduct preliminary fault diagnostic on a DF (CIO) System (Tech of the Watch)	
E027	Configure a DF (CIO) System to meet an operational requirement	
E028	Configure/reconfigure DF (CIO) System software	
E029	Diagnose a DF (CIO) System fault to the Lowest Repairable Unit	
	DRMIS	
	As per DRMIS QSP 119097.	

ANNEX C – QS SCALAR



ANNEX D – RECORD OF DECISIONS QSWB

1. This record summarizes the conduct of the QSWB including major activities, important discussions and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

Dates:

12 Apr – 30 Apr 10

Location:

CFFS(E) Bldg N92 Rm 300

Board Members:

SURNAME	INITIALS	RANK	MOSID	UNIT
STONIER	P.C.	CPO1	00119	D MAR PERS 3-2-3
MAGEE	M.G.	CPO1	00119	DMTE
DEMOE	J.E.	PO1	00118	CFFS(E)
OICKLE	B.W.	PO1	00118	CFNES
GAUTHIER	J.H.	PO2	00118	HMCS ALGONQUIN
FURMAN	J.R.D.	MS	00118	HMCS OTTAWA
MURRAY	D.R.	CIV	EL-07	CFNES ERT DIV
VAN DUSEN	D.G.	MS	00299	HMCS CALGARY

Board Objective:

To create the Job Qualification for the W Eng Radar System Maintainer.

Discussions / Decisions:

Day 1 – 12 Apr 10

1. There were introductions. CPO1 Stonier discussed Job-Based Specification, Job Qualification, the training modules of the new W ENG Tech Occupation. Training philosophy is new OS is not expected to be an expert, but building on his trade knowledge as OJT onboard the ship. Second OJT grants the LS rank qualification. There will b a board at the end of the second OJT, to ensure standard being met throughout the fleet. The philosophy is to have the Trainee get into the books and learn on the job. Give them the academic training, and 80% solution for application training. The idea is to train them to become independent thinkers. New philosophy is "Conceptual knowledge" cs "Detailed knowledge." Want to train to be able to do the job. The course is essentially an introduction to each system. The benefits of MPT training vs working on the real gear were discussed. HAZMAT training is not an individual skill, it's a team skill: donning a Dreger is taught on NETP.

2. CPO1 Magee discussed DMTE point of view. DMTE requires finished product from this board. By 15 Dec, need to produce a trainable QSP. 1 Apr 2011 is the startup date for the first serial for the Jr W Eng Tech Module 1.

3. Dave Murray discussed there will be 4 semesters, the junior is a 12 week and 9 week, throughput will be faster than previous training. All Trainees are getting all the same training. CPO1 Stonier mentioned that there are 5 system maintainers in W ENG TECH. Trainees will have 3 choices for which they want to specialize in. Will try to give them what they want. The idea is to minimize backlash from Trainees not getting the choice they want. Cross-training will be much easier, because all have common training. PO1 Demoe mentioned that a big dissatisfier to new Trainees is not getting to sea in the initial years. Under the new system, will get to sea more quickly.

4. Lt(N) Giroux gave the TDO Intro Brief. He discussed the Mandate, Board Agenda, Schedule, Roles, and responsibilities, Board Conduct, Record of Proceedings, and CFITES. The Mandate is to "create the QSP for the W Eng Tech Radar Formal Training that results in properly qualified personnel capable of carrying out duties commensurate with their rank and MOSID requirements."

5. PO1 Pickle inquired about Chapter 3 of the QSP, saying he would like it to be discussed during this board. CPO1 Magee said that it should be a complete QSP by the time this board is complete, including Chapter 3 WRT the assessment criteria.

6. Lt(N) Giroux demonstrated the Naval Training System Wiki, in which all data collection will be documented and then transferred to QSP template. This is the first time that this technology has been used for a QSP.

7. CPO1 Magee discussed scalars, which take all tasks and puts them in a logical sequence for training. He also discussed the new trade progression system. CPO1 Stonier said the idea behind QRA training is to train the individual for the specific job they

require, as opposed to training everyone for every job they may hold throughout their career in the trade.

8. In the afternoon, the board reviewed the LS Qualification of the QSP and the various pieces of equipment.

Day 2 – 13 Apr 10

1. Reviewed ROP.

2. Received a briefing about the Wiki site, and how it can be utilized to conduct the boad more efficiently.

3. In the morning, we discussed scalars, and as the job is similar to the LS Rank Qual, the board adopted the scalar that was designed and used for the LS Rank Qual QSP to create the performance objectives.

4. The board started and completed Chapter 2 Performance Objectives using the Wiki to collect data and conduct concurrent activity.

Day 3 – 13 Apr 10

5. Reviewed ROP.

6. Reviewed Chapter 4. After discussion in EO 001, we decided we needed master lesson plan to determine if time allotted to current course was calid with our philosophy of training.

7. The board began reviewing Pos 1 and 4 using the Master Lesson Plans. Working in the wiki, the board continued to streamline training based on new philosophy.

Day 4 – 14 Apr 10

8. Reviewed ROP.

9. Practice faultfinding on equipment will always be conducted in teams of two. Therefore, in order to get a class of eight students through one practice fault four periods are allotted.

10. The board discussed CCS training and decided that the focus for Radar System Maintainers would be CCS architecture vice maintenance.

11. The board discussed the decreased length in training vs the ship's expectation of the member. The idea is to teach the students to think on their own, vice teach each piece of equipment in detail.

12. The board continued working on Chapter 4 throughout the da. PO 001 and 002 were completed and reviewed by the board, and are ready to transfer to the template.

Day 5 – 16 Apr 10

13. Reviewed ROP.

14. The board recommends that the CCS block diagram be included in the Jr W Eng Tech JQ Mod 3.

15. The board discussed PO 007, CCS. It was decided that a CCS management phase was not necessary.

16. The board worked to design EO 007.01 and 007.02. Discussed the possibility of giving the PO to all occupations, decision was withheld until this Radar document is complete.

17. The board began discussing Chapter 3 and marking criteria. The difference between ECs and Critical ECs were discussed. Discussion centered on the inclusion of EC marks and the final overall PO/PC mark. The TDO was brought in to clarify. Following the discussion, the board decided unanimously that it was critical to the correct assessment of the Trainee that ECs would count towards the final mark of each PO. Homework assignments, labs, and worksheets will also contribute to the final mark The final PC will be worth no less than 75% of the final mark.

18. The board finished Chapter 3 and began to work on Annex F. This will be a common scoring matrix for EC/PCs.

Day 7 – 20 Apr 10

19. Reviewed ROP.

20. The board designed the marking scheme for Annex F. Four areas for assessment were inserted into the marking sheet for a total of 40 points, 5 point swere given for time, 5 points were given for finding the fault. At this point, the board examined the marking scheme and determined that it was possible for someone to find the fault and still fail. The pass mark was adjusted to 60% to reflect the new marking philosophy. It was felt that the new marking scheme had more granularity than previous marking schemes and that the standard was not being lowered, but the student's ability will be more accurately assessed. This will also address recent shortcomings noted by the training establishment wherein the candidates;

performance at sea was not accurately reflected by their mark. i.e. A student received very high marks in faultfinding, but their faultfinding performance on the ship was substandard.

21. The board began reviewing and amending the reference Annex and applicable references in the POs and EOs.

Day 8 – 21 Apr 10

22. Reviewed ROP.

23. The board reviewed and amended the reference Annex and applicable references in the Pos and EOs. The board began the arduous task of building the teaching point tables in each EO

Day 9 – 22 Apr 10

24. Reviewed ROP.

25. The board completed building the teaching point tables in each EO. The board revisited Annex K and using input obtained from CFNES and CFFSE equipment instructors adjusted the scoring matrix.

26. The board discussed W Eng Tech System Maintainer Common Academics. These Common Academics had been previously outlined during the W Eng Tech LS-RQ QSP Board in Mar 10. Mr. Murray made several suggestions to the draft that were accepted by the board and following secretarial acceptance from the previous board members will be implemented in the W Eng System Maintainer Common Academics QSP to be reviewed at a later QSP Board. It was agreed that CFNES E & T Div would populate the teaching point table for each of the predetermined EDOs.

Day 10 – 23 Apr 10

27. Reviewed ROP.

28. The board reviewed the QSP in its entirety with the TDO, Capt. Normore and made minor changes throughout the QSP to ensure its compliance with MARCORDs, CFITES, and other governing documents. The board completed Lessons Learned and Guidance to the Instructors sections of the QSP, as well as follow-up action. Board Chairman thanked all participants for their expertise and for expressing themselves in a manner that aided the creation of the QSP.

Lessons Learned

29. When using the QSP Wiki, it is imperative that all SMEs use the exact same format. This will eliminate multiple corrections on final document review.

30. The Wiki is incapable of producing tables that are easily edited. Therefore, you must cut and paste content into the final document. It was found that the Wiki's primary strength is for producing content concurrently, as opposed to formatting.

31. The board noticed that it was sometimes necessary to double-save the Wiki in order to save documents

.

32. All supporting documentation, i.e. Master Lesson Plans, timing information, and

33. Reference lists should be available to the board.

Guidance to Instructional Designers.

34. Ideally, all practice fault-finding is conducted in pairs. To conduct on practice fault, the board allotted hours for a course of 8. If 20 periods are allotted for practice faults, the board intention is to have 5 practice faults for that period. If time allows, the instructors are free to insert additional faults, or consider individual faults.

35. The level of detail taught on any specific PO should e commensurate with the time allotted.

Follow-Up Action :

36. DMTE to create and insert task lists. On completion, DMTE to distribute completed QSP to board members, TDO, and supervisor for approval.

Other Considerations (If applicable) :

N/A.

TRAINING PLAN ANNEXES

ANNEX E – RESOURCES, EQUIPMENT, SUPPLIES, AND ANCILLARIES

1. The following resources, equipment, supplies, and other elements are required in support of training. Resources are described in terms of formed organizations and are to arrive equipped with all personnel, vehicles, weapons, communications, and ancillary equipment required for conduct and integral support of operations described in the POs.

2. Facilities/Trainers

SER	DESCRIPTION		QTY	REMARKS
a.				EO001.02 and 009.11 Security Classification required
b.				
c.				
NOTES:				

3. Ammunition

SER	LOG GUIDE	DESCRIPTION		QTY	REMARKS
a.					
b.					
c.					
NOTES:					

4. Weapons

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

5. Specialized/Technical Equipment

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

6. Transport/Vehicles

SER		NSN	DESCRIPTION		QTY	REMARKS
a.		XXX-XX-XXX-XXXX				
b.		XXX-XX-XXX-XXXX				
c.		XXX-XX-XXX-XXXX				
	NOTES:					

7. Communication Equipment/Frequencies

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

8. Training Aids

SER	DESCRIPTION		QTY	REMARKS
a.				

b.				
c.				
NOTES:				

9. POL Products

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

10. General Purpose Stores

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

11. Stationery

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

12. Personal Equipment

All students will have in their possession the personal equipment detailed in the course joining instructions. All personal equipment to be assigned to the students are accounted for in the resource tables above.

ANNEX F – COSTING INFORMATION

Estimates only, based on CF Cost Factors Manual

cost-factors-manual (mil.ca)

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course: EXAMPLE		Serial:				
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Airfare (Return)	14	$1,000.0		$14,000	$14,000	$70,000.00
R&Q	14	$36.0	20	$10,080	$10,080	$50,400.00
Incidentals	14	$17.5	20	$4,900	$4,900	$24,500.00
Subtotal				**$28,980**	**$28,980**	**$144,900.00**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	16	$50.0	1	$800	$800	$4,000.00
Classroom (square meters)	30	$13.00	15	…	…	…
Computer requirements	16	$1,000.0	1	$16,000	$0	$16,000.00
IT Equip (Projector, Smart board etc.)	1	$6,000.0	1	$6,000	$0	$6,000.00
DWAN connection (Set up)	1	$5,000.0	1	$5,000	$0	$5,000.00
Link Toolkit (on DVD)	15	$1.0	1	$15	$15	$75
Instructional Materials (Printed / Handouts etc.)	15	$30.0	1	$450	$450	$2,250.00
Subtotal				**$28,265**	**$1,265**	**$33,325.00**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
Instructor (PO1/WO) (incremental)2	2	$223.0	20	$8,920	$8,920	$44,600.00
Incremental Civilian Instructor		XXX				

		subtotal		**$8,920**	**$8,920**	**$44,600.00**
	Total Estimated Costs			**$66,165**	**$39,165**	**$222,825**

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

1. This annex is to be used to properly schedule all required classrooms, trainers, ships, personnel, etc. for the duration of the course. Admin and PT periods are accounted for IAW NPTORD 4500-22 Annex B (QSP Instructions).

2. The tables are completed in the preferred sequence for lessons and resources. Special instructions to schedulers shall be included as "Notes" following the applicable PO or Module.

3. The table below is available to use as a template for calculations.

Allocation of ATT	**Periods**
Actual Training Time	365
ATT Sub-total	**446**
Allocation of BAT, TAT, TTT, PT (Course > 31 days)	
Base Administrative Time (BAT)	7
Training Administrative Time (TAT)	26
Training Travel Time (TTT)	0.00
Physical Fitness (PT)	39
Sub-total	**72**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **Total mins/400 = # days** (rounded up)	
Total Training Days	**65**

OPTIMUM TRAINING SCHEDULE

Week 1 Time	Day 1	Day 2	Day 3	Day4	Day 5
Period 1 **0800-0850**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 2 **0900-0950**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 3 **1000-1050**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 4 **1100-1150**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 5 **1300-1350**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 6 **1400-1450**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 7	**EO/EC /PC/Admin** Title(s) / Activity				

Week 1 Time	Day 1	Day 2	Day 3	Day4	Day 5
1500-1550					
Period 8 **1600-1650**	**EO/EC /PC/Admin** Title(s) / Activity				

ANNEX H – ASSESSMENT PLAN

EC / PC	Scope Covered	Purpose	How	When	Resources	Pass Mark	Impact
EC 001.03	EO 001.03	To evaluate the Trainees ability to repair the Long Range radar system	Practical	O/C of EO 001.03	Long Range radar system	60%	Re-attempt TRB
PC 001	EO 001.01 Thru EO 001.03	To evaluate the Trainees ability to maintain the 3D radar system	Practical	O/C of EO 001.06	Long Range radar system	60%	Re-attempt TRB
EC 002.03	EO 002.03	To evaluate the Trainees ability to maintain the 2D radar system	Practical	O/C of EO 002.01	Medium Range radar system	60%	Re-attempt TRB
PC 002	EO 002.01 Thru EO 002.03	To evaluate the Trainees ability to maintain the Medium Range radar	Practical	O/C of EO 002.07	Medium Range radar system	60%	Re-attempt TRB
EC 003.01	EO 003.01	To evaluate the Trainees ability to maintain the IFF System	Practical	O/C of EO 003.01	IFF System	60%	Re-attempt TRB

EC 003.02	EO 003.02	To evaluate the Trainees ability to maintain the IFF System	Practical	O/C of EO 003.02	IFF System	60%	Re-attempt TRB
EC 003.03	EO 003.03	To evaluate the Trainees ability to maintain the IFF System	Practical	O/C of EO 003.03	IFF System	60%	Re-attempt TRB
PC 003	EO 003.01 Thru EO 003.03	To evaluate the Trainees ability to maintain the IFF System	Practical	O/C of EO 005.03	IFF System	60%	Re-attempt TRB
EC 004.03	EO 004.03	To evaluate the Trainees ability to maintain the ESM System	Practical	O/C of EO 004.03	ELISRA	60%	Re-attempt TRB
PC 004	EO 004.01 Thru EO 004.03	To evaluate the Trainees ability to maintain the ESM System	Practical	O/C of EO 004.03	ELISRA	60%	Re-attempt TRB
PC 005	EO 005.01 Thru EO 005.03	To evaluate the Trainees ability to maintain the Sea Search System	Practical	O/C of EO 005.03	Sea Search System	60%	Re-attempt TRB
PC 006	EO 006.01 Thru EO 006.03	To evaluate the Trainees ability to maintain the CIS System	Practical	O/C of EO 006.03	AN/SRD 501	60%	Re-attempt TRB
PC 007	EO 007.01 And EO 007.02	To evaluate the Trainees ability to maintain the CMS	Practical	O/C of EO 007.02	CMS	60%	Re-attempt TRB

PC 008		As Per DRMIS QSP MOD 2 (119097)					

PERFORMANCE CHECK/ENABLING CHECK INSTRUCTIONS

PART I – ADMINISTRATIVE INSTRUCTIONS

This is a generic list for administering a PC/EC

1. The assessor shall:
    a. Ensure all publications, test equipment, and equipment are serviceable or Trainee informed of deficiencies;
    b. Ensure the Trainees observe all safety precautions;
    c. Test Trainees individually;
    d. Supervise and provide physical assistance if required;
    e. Provide a pre and post assessment waiting area; and
    f. Review Annex K with Trainee.

2. If, after allotted time, the Trainee is showing sufficient progress, the Assessor may allot 15 additional minutes to complete the PC/EC with the loss of two points. For each additional 5 minutes after the initial 15 minutes will result in two points being lost. The maximum extra time granted will be 30 minutes. If the student does not find the fault after the additional 30 minutes, the instructor will stop the PC/EC and conduct the final mark.

3. The Assessor is to ensure that the best use of time allotted for PCs/ECs is made by facilitating the PC/EC process. For example:
    a. If the Trainee is about to perform a logical but time-consuming procedure the Assessor should supply the results and have the Trainee resume without penalty to the student. However, if the action is not justified by the Trainee it is considered an incorrect procedure; or
    b. After a maximum of 10 minutes inactivity the Assessor shall provide assistance.

4. Pass mark is 60%.

Part II – INSTRUCTIONS GIVEN TO TRAINEE

1. The Trainees are given the following:

    a. References;

    b. Course notes;

    c. Job sheets;

    d. Work sheets;

    e. Projects; and

    f. Equipment and/or Applicable Training Simulator.

2. IAW applicable references and PC/EC list, the Trainee shall:

    a. Power up;

    b. Conduct system functional checks;

    c. Load software or crypto as required;

    d. Localize fault to LRU;

    e. Repair;

    f. Verify operational readiness; and

    g. Power down.

3. Prior to performing a test or measurement the Trainee shall inform the Assessor of their intention.

4. IAW applicable references and PC/EC list, you shall complete the PC/EC:

   a. Within allotted time IAW Chapter 4;

   b. Individually;

   c. With supervision; and

   d. Physical assistance as required.

5. IAW applicable references, the Trainee shall observe all safety precautions. A safety infraction or violation will result in instructor intervention and the instructor may terminate PC/EC based on the severity of the violation.

6. Pass mark is 60%.
7. Does the Trainee have any questions?

Trainee Signature:_____ Date:_____

Instructor Signature:_____ Date:_____

ANNEX I – ASSESSMENT CHECKLISTS From Old Format

PC/EC EVALUATION SHEET			
Equipment:		**Name:**	
PO Number:	**Start:**	**Stop:**	**Total:**
Course:			
Note: When allocating a score use only whole numbers; and Circle the appropriate number in score column			

Initialization (10%)		
Score	**Justification**	**Remarks**
1-2	Did not follow correct flash up procedure. Flash up faults not recognized	
3-4	Followed procedure but flash up faults not recognized	
5-6	Followed flash up procedure. After some hesitation, corrected flash up faults	
7-8	Followed Flash Up procedure. Recognized and corrected flash up faults	
9-10	Confidently Flashed Up, efficiently corrected flash up faults.	

Use of Test Equipment (10%)		
Score	**Justification**	**Remarks**
1-2	Did Not select appropriate TE for Fault/ Did not demonstrate correct TE operating procedures	
3-4	Hesitant with selection of TE/Unsure of TE set up	
5-6	Selected correct TE/ Demonstrate weak operating procedures	
7-8	Select Correct TE/Properly utilized TE	
9-10	Selected Correct TE considering best options/Demonstrated thorough knowledge of operation	

Fault Finding Procedures (30%)		
Score	**Justification**	**Remarks**
1-2	Hesitant to begin Fault Finding./Did not recognize fault indications	
3-4	Did not correctly interpret Fault Indications; BITE; Test Equipment Readings; Took the wrong path and did not recover.	
5-6	Minor inconsistencies in interpreting fault indications. May have taken wrong path but	

	recovered. Employed logical faultfinding.	
7-8	Recognized and Correctly interpreted fault indications. Employed Logical Faultfinding.	
9-10	Efficiently interpreted Fault Indications and evaluated all symptoms. Found fault with minimum number of steps.	
System Knowledge (30%)		
Score	**Justification**	**Remarks**
1 - 2	System knowledge did not support faultfinding. Could not answer questions from instructor. Could not properly use supporting documents	
3 - 4	System knowledge weak. Could not answer all questions. Struggled with documentation to support faultfinding. Could not fully explain fault	
5 - 6	System knowledge weak in some areas. Answered most questions from instructor and gave general explanations regarding the fault.	
7 - 8	Good system knowledge. Use of supporting documentation aided in fault finding.	

	Explained fault clearly.	
9 - 10	Detailed system knowledge. Used documentation efficiently to find fault. Gave detailed explanation of fault	
Timing (10%)		
Score	**Justification**	**Remarks**
10	Found Fault in allotted time	
8	Found Fault with an extra 15 minutes	
6	Found Fault with an extra 20 Minutes	
4	Found Fault with an extra 25 Minutes	
2	Found Fault with an extra 30 Minutes	
Find the Fault (10%)		
Score	**Justification**	**Remarks**
10	Fault Corrected	

Score	**Weighting**	**Total**
Initialization	X 1	%
Test Equipment	X 1	%
Fault Finding	X 3	%
System Knowledge	X 3	%
Timing	X 1	%
Finding the Fault	X 1	%
	TOTAL	%

Assessor Signature:	**Trainee Signature:**

New format

PERFORMANCE/ENABLING CHECK

PC/EC Time: ### mins (concurrent/per trainee) + ## min debrief **ASSESSMENT ACTIVITY**: (Brief description of the assessment activity that students will do IOT prove they have met the Standard of the PO/EO). **Pass/fail criteria**: (identify success criteria) **ASSESSMENT RESOURCES:** • (Bulleted list of everything given to trainees for PC/EC) **IMPACT OF FAILURE:** Remedial Training / Re-test

INVIGILATOR NOTES:

(Course Name) – PC/EC XXX.XX Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

1. PASS / FAIL CRITERIA: Identify success criteria.

2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	**PASS**	**FAIL**	**COMMENTS**
1. Criteria			
(Success criteria taken from PO/EO standard)			
2.			
ADDITIONAL COMMENTS			
STRENGTHS:	**AREAS FOR DEVELOPMENT:**		
THE TRAINEE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			

OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

ANNEX J – ON JOB PERFORMANCE RECORD (OJPR)

(Certificate of Qualification)

Version	Revision Date	Remarks
OJPR Version 1.0	DD / MM/ YYYY	New OJPR

Name and Initials	Rank	Service Number

Date Training Started	DD MMM YYYY
Date of Completion	DD MMM YYYY

THE OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE OJPR WILL BE RETURNED FOR RE-SUBMISSION.

INSTRUCTIONS

1. One copy of this OJPR, hereinafter referred to as the master copy, shall be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. The Training Establishment (TE) shall make a duplicate copy for retention by the trainee for the purpose of recording and reviewing individual progress. The TE for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

2. As the trainee completes each Performance or Enabling Check to the required standard, IAW the associated checklist in ANNEX I, the Training PO or Officer will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and inform the individual of the action taken.

3. It may not be possible to conduct OJT for some POs if, for example, the necessary equipment is not fitted on a ship, is not on issue to the unit, or is otherwise unavailable. If the training PO or Officer encounters any of these issues they are directed to contact Standards at the appropriate TE for guidance.

4. If an individual demonstrates the ability to perform additional duties not specified in this QSP, an appropriate entry shall be made under "Additional Duties" at the end of the OJPR.

PROGRESS REVIEW

5. Progress review shall be carried out quarterly by the Divisional Officer and Section Head (NCMs) or Training Officer (Officers).

Example OJPR record:

PO/EO	Description	Remarks	Rank/Name/ Signature	Date	OPI and Unit
PC/EC XXX.XX				DD/MM/YYYY	

OJPR Reviewed By OJT Supervisor (Rank, Position, Name)	Signature	Date
Section Head (NCMs) or HOD (Officers)		

TRAINING COMPLETION CERTIFICATE

Date training completed:	
Div Officer (NCMs) or CO (Officers)	Date

CERTIFICATION OF QUALIFICATION

I certify that _____ has completed the OJT to the standards stated in the Name of QSP.

TE Cmdt/CO	Date

ANNEX K – TT PLAN

PO / EO	Objective	EO	EC	PC	PO
PO 001	Maintain the Long Range Radar System	61	8	8	77
EO 001.01	Describe the Long Range Radar	18			
EO 001.02	Initialize the Long Range Radar	7			
EO 001.03	Repair the Long Range Radar System	35	8	8	
EO 001.04	Perform Preventive Maintenance (PM) on the Long Range Radar System	8			
PO 002	Maintain the Medium Range Radar System	67	8	8	83
EO 002.01	Describe the Medium Range Radar System	2			
EO 002.02	Initialize the Medium Range Radar System	3			
EO 002.03	Repair the Medium Range Radar System	48	8	8	
EO 002.04	Perform Preventive Maintenance (PM) on the Medium Range Radar System	6			
PO 003	Maintain Identification Friend or Foe System	67	3	8	78
EO 003.01	Describe the IFF System	8	1		
EO 003.02	Initialize the IFF System	13	1		
EO 003.03	Repair the Transponder Assembly	33	1	8	
EO 003.04	Perform PM on the IFF System	13			
PO 004	Maintain Electronic Support Measures System	47	8	8	63
EO 004.01	Describe the Electronic Support Measures system	7			
EO 004.03	Repair the ESM System	33	8	8	
PO 005	Maintain Electronic Support Measures System (Sea Search)	44		8	52
EO 005.01	Describe the Operation of the ESM System AN/ULR 501 (Sea Search)	10			
EO 005.02	Initialize the ESM System AN/ULR 501 (Sea Search)	6			
EO 005.03	Maintain the ESM System AN/ULR 501 (Sea Search)	28		8	
PO 006	Maintain Communication Intercept System	44		8	52
EO 006.01	Describe the Operation of the Communication Intercept System	7			
EO 006.02	Initialize the Communication Intercept System	7			
EO 006.03	Maintain the Communication Intercept System	30		8	
PO 007	Maintain the Combat Management System (CMS 330)	30		5	35
EO 007.01	Describe the Combat Management System (CMS 330)	10			

EO 007.03	Maintain the Combat Management System (CMS 330)	10		5	
PO 008	Utilize DRMIS Portal as a Maintenance Supervisor IAW QSP 119097	5		1	6
EO 008.01	Utilize DRMIS Portal as a Maintenance Supervisor IAW QSP 119097	5		1	

ANNEX L – RECORD OF DECISIONS TPWB

1. This record summarizes the conduct of the TPWB including major activities, important discussions and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

Title				
Location of Board				
Dates				
Board Composition				
	Surname	Initials	Rank	Unit
Co-Chair (content)				
Co-Chair (process)				
NFS				
SME				
SME				
SME				
SME				
SME/Secretary				
Advisor				

Day #	Details	Action
	The following information on target population was compiled during the initial steps of the TPWB to give context for design and development: (rank, experience, entry level education, etc.).	
1		
2		

RECORD OF AMENDMENT & AMPLIFICATION

1. The record of amendment and amplification will be filled in whenever a change to the TP is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted TP Amendment Form, thus ensuring the QSP has the most relevant information.

Version:	**X.X**
Date:	**Who Initiated Change:**
Requirement for the Amendment:	
Description of Changes:	
Change Tracking number:	

Trg Mgr:

ANNEX M – ACRONYMS AND ABBREVIATIONS

1. This annex is to be utilized only to identify uncommon abbreviations used within this QSP.

ACRONYM / ABBREVIATION	TERM/DEFINITION
BITE	Built In Test Equipment
CCS	Command and Control Systems
CSAT	Combat System Alignment Test
CSIT	Combat System Integration Test
CSOT	Combat System Operability Test
DWAN	Defence Wide Area Network
LAN	Local Area Network
LRU	Lowest Replaceable Unit
MPT	Maintenance Procedural Trainer - Can be any form of computer based training, table top faults, or any other form of fault finding practice necessary.
OFL	Offline BITE
POFA	Programmed Operational Functional Analysis
SHIPLAN	Ship Local Area Network
UPS	Uninterruptable Power Supply